Living Architecture

UNIVERSITÀ DEGLI STUDI DI TRENTO

CIBIO

Quantifying Mechanisms of Bacterial Phosphate Economy for Synthetic Apps

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The project has received funding from the European Union's Horizon 2020 Research and Innovation Programme under Grant Agreement no 686585.

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EU Horizon 2020 LIAR – Living Architecture

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LIAR - Living Architecture

EU Horizon 2020

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Building next-generation, selectively-programmable bioreactors.

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LIAR - Living Architecture

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GOAL: improving the water quality

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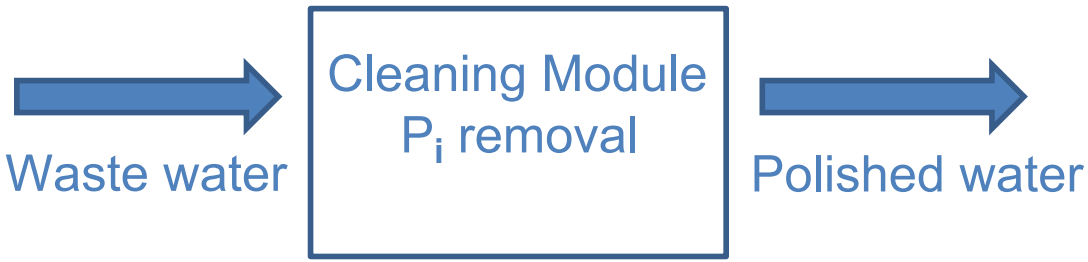

SOLUTION: removing inorganic phosphate (P_i)

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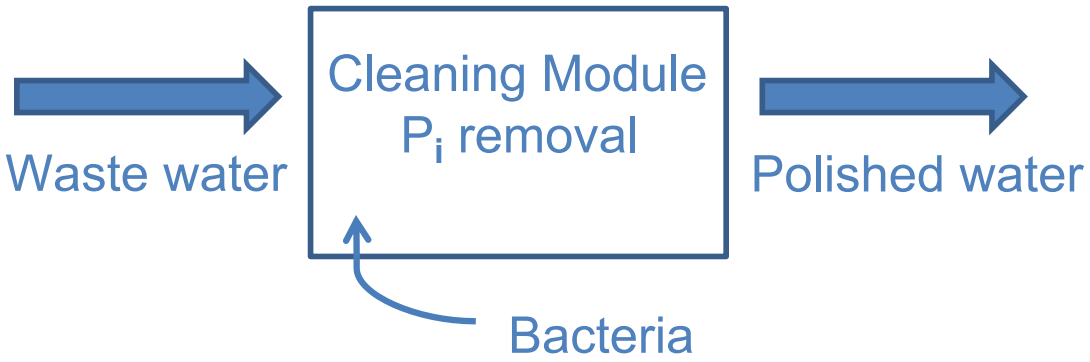
GOAL: improving the water quality



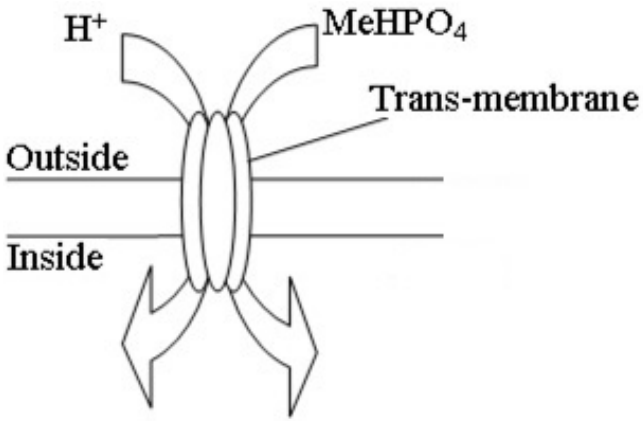
SOLUTION: removing inorganic phosphate (P_i)

Inorganic Phosphate Transport Systems

Inorganic Phosphate (P_i) transport

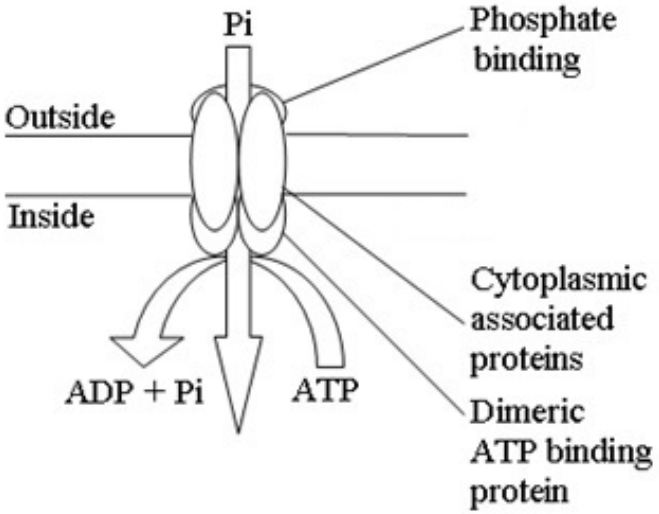
Pit - Low Affinity **P**hosphate **I**norganic **T**ransport

- high external concentration of P_i



Pst - High Affinity **P**hosphate **S**pecific **T**ransport

- low external concentration of P_i



Pst and PHO Regulon

How does *E. coli* sense the environmental stimuli?

Pst and PHO Regulon

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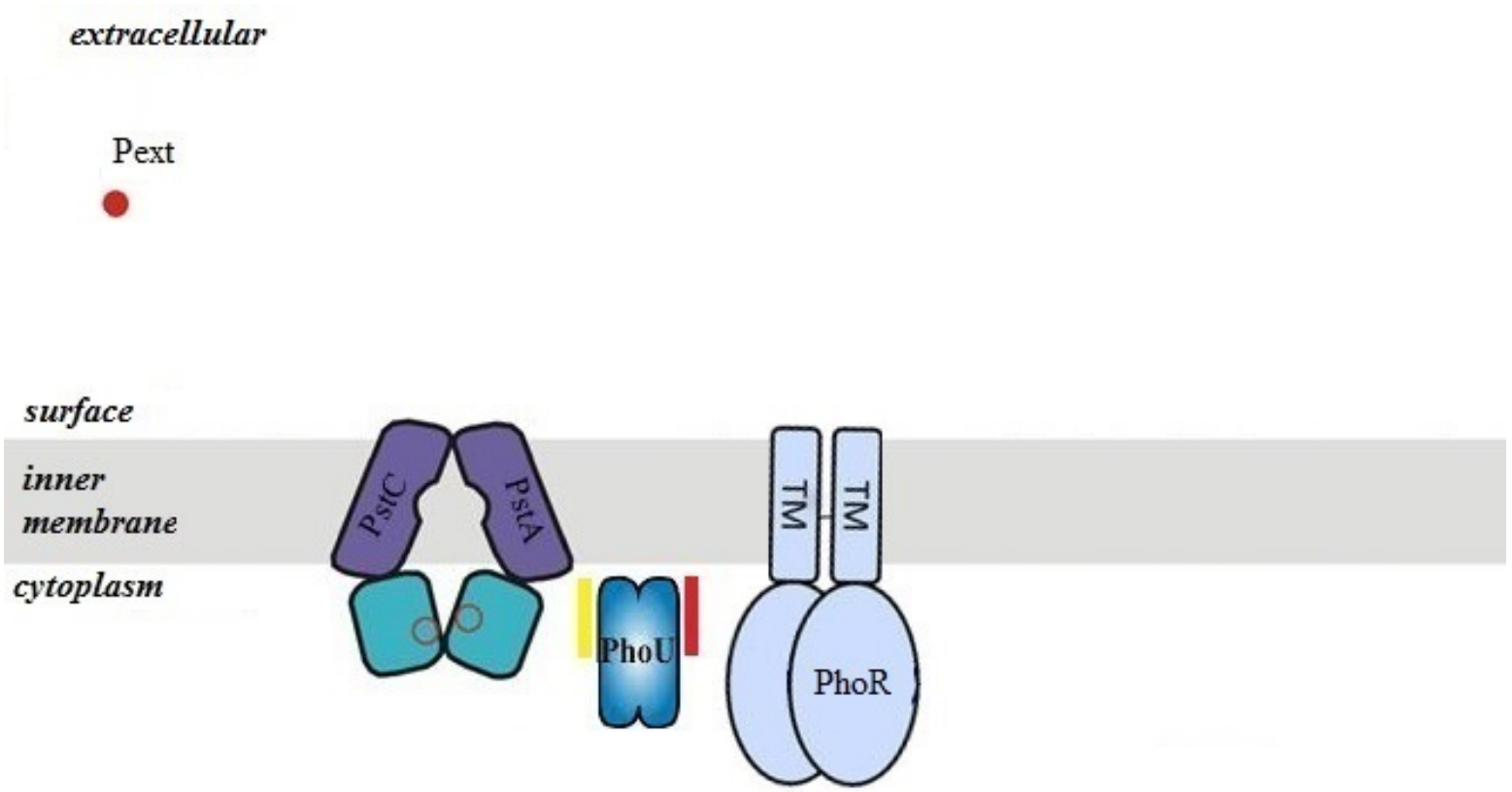
- Pst system is responsible for sensing the environmental changes and importing the P_i into the cell.

Pst and PHO Regulon

How does *E. coli* sense the environmental stimuli?

- Pst system is responsible for sensing the environmental changes and importing the P_i into the cell.
- Pst relays the signal to a two-component system (TCS) to adapt to current conditions by regulating the expression of Phosphate (PHO) regulon genes.

PHO Regulon when Pext is limited



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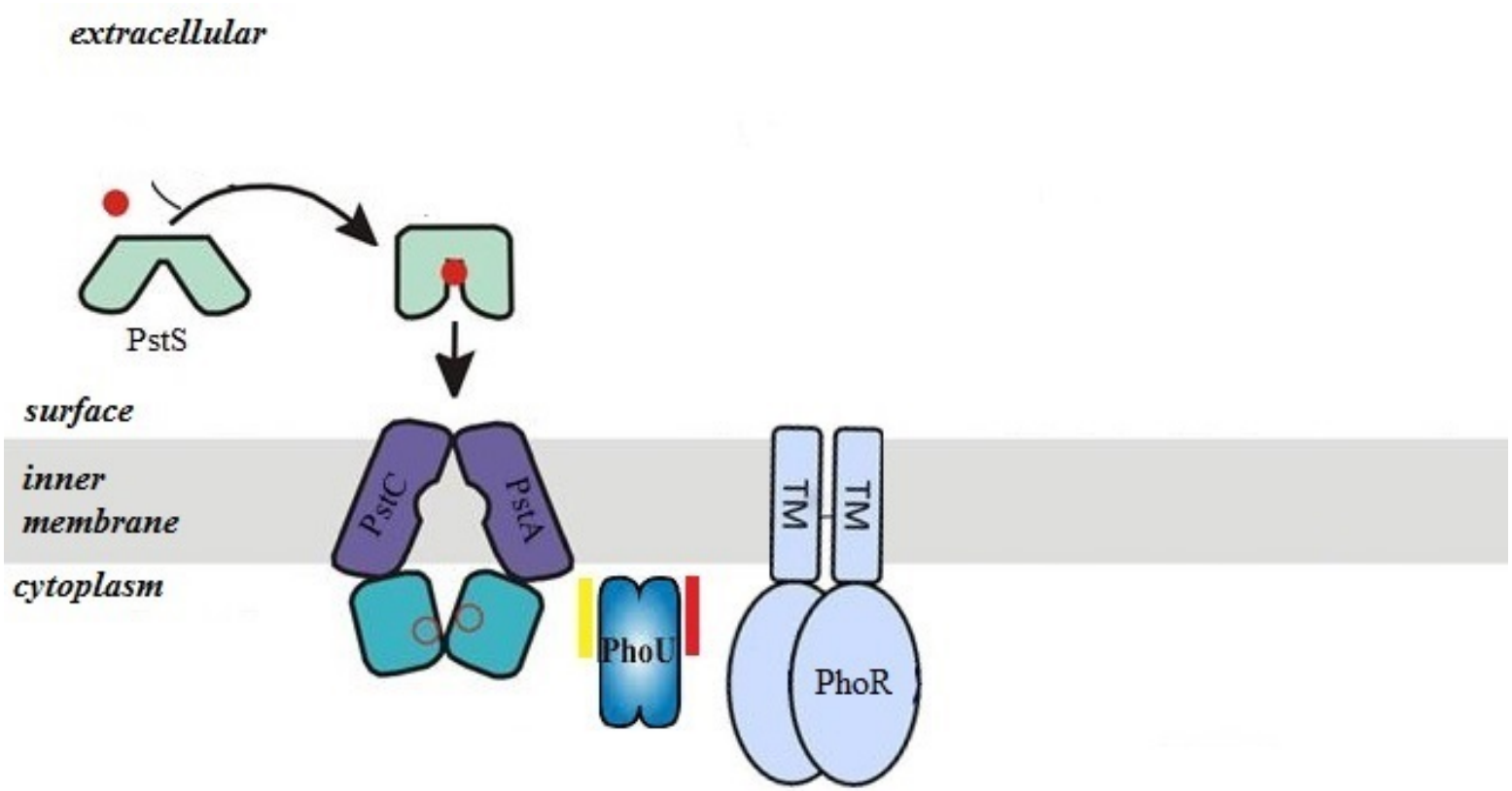

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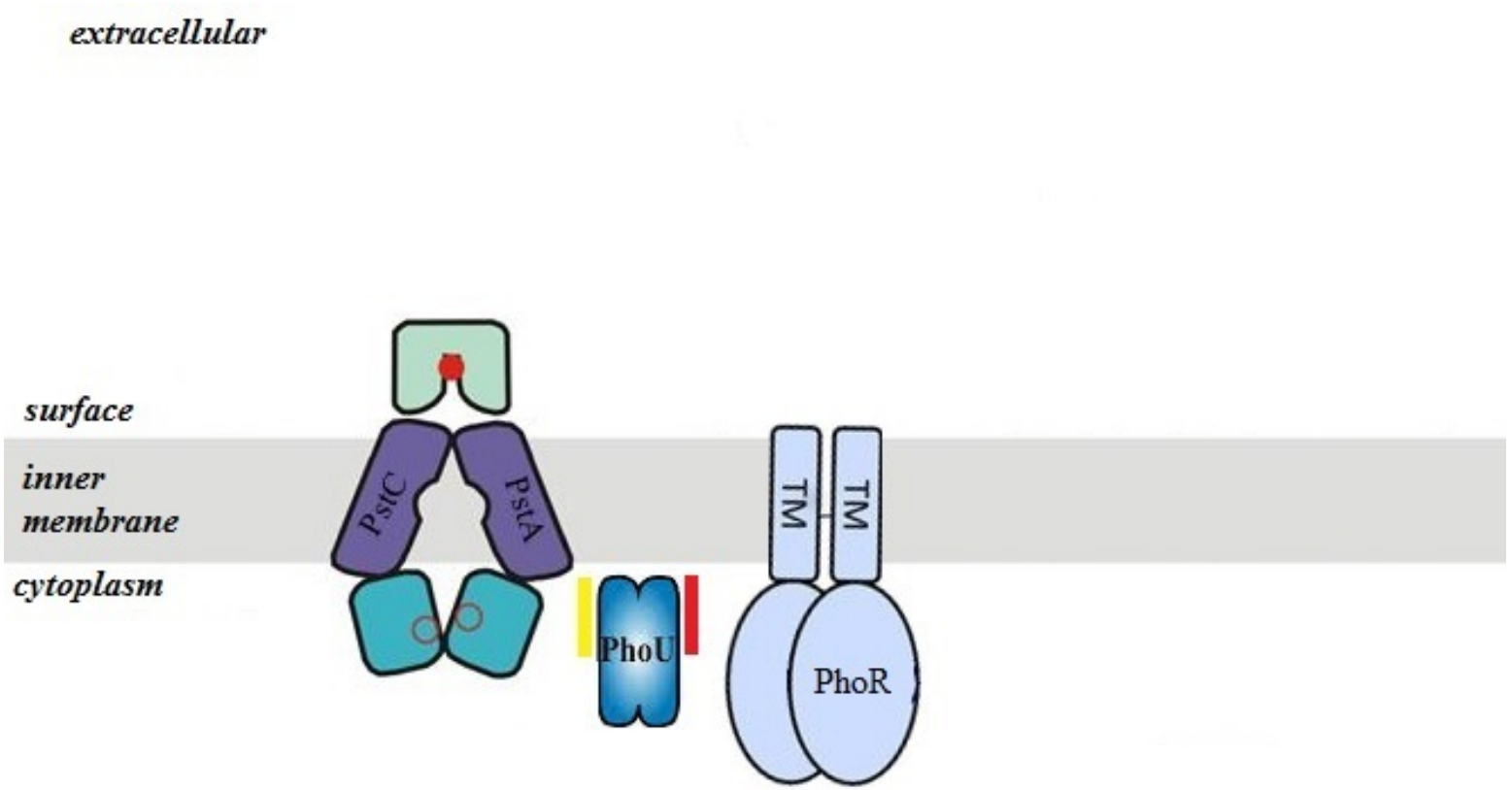

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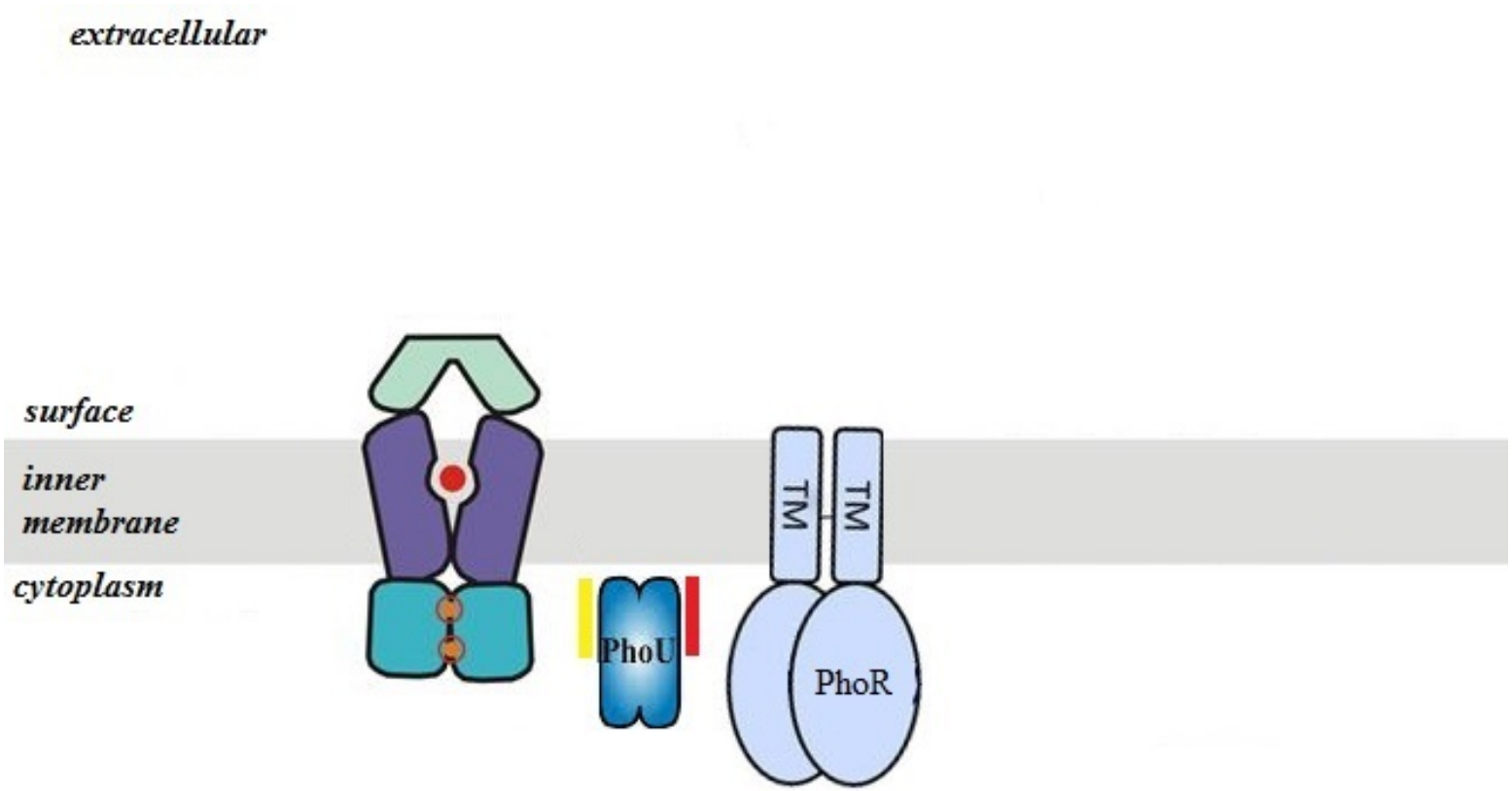

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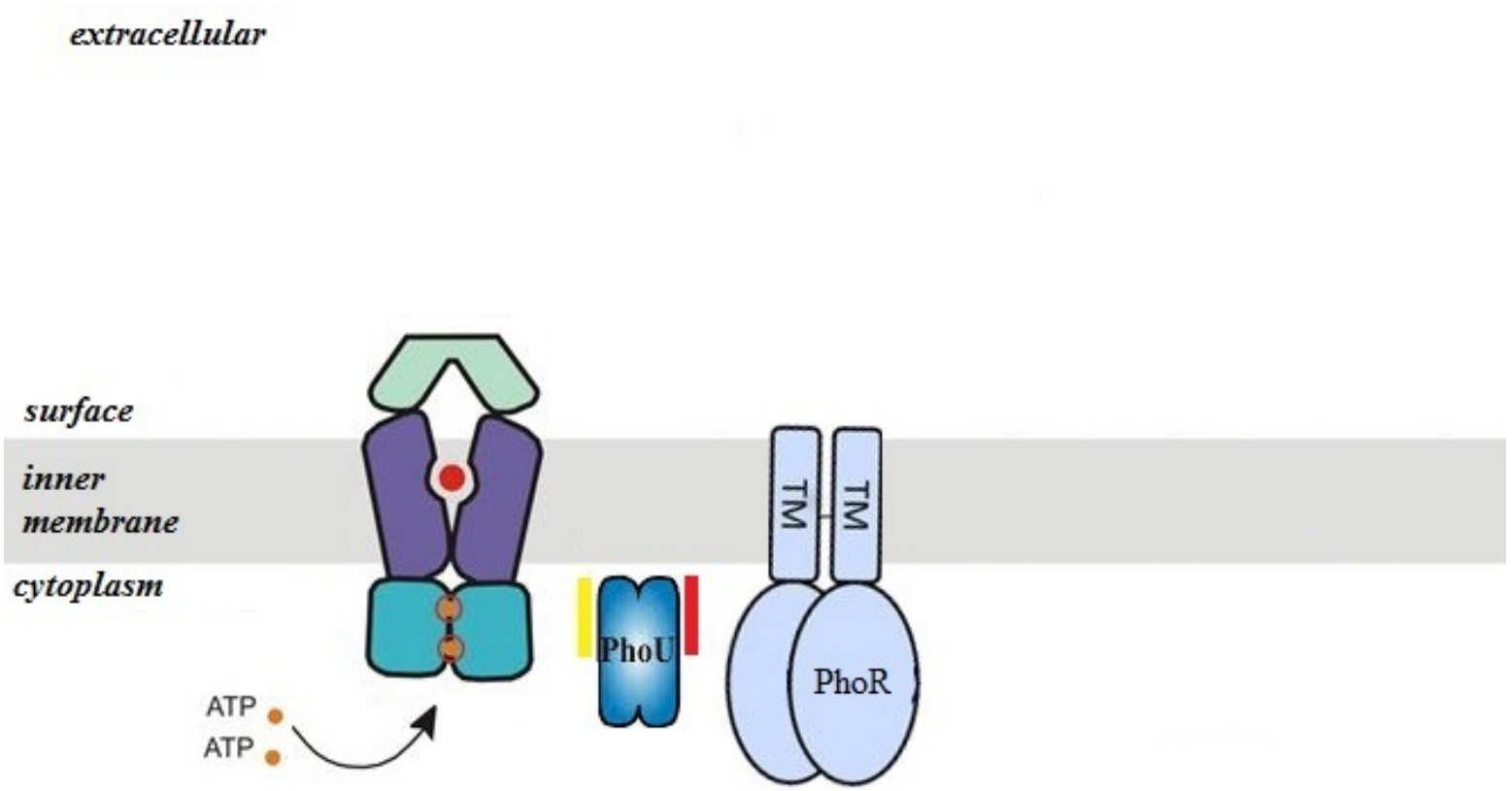

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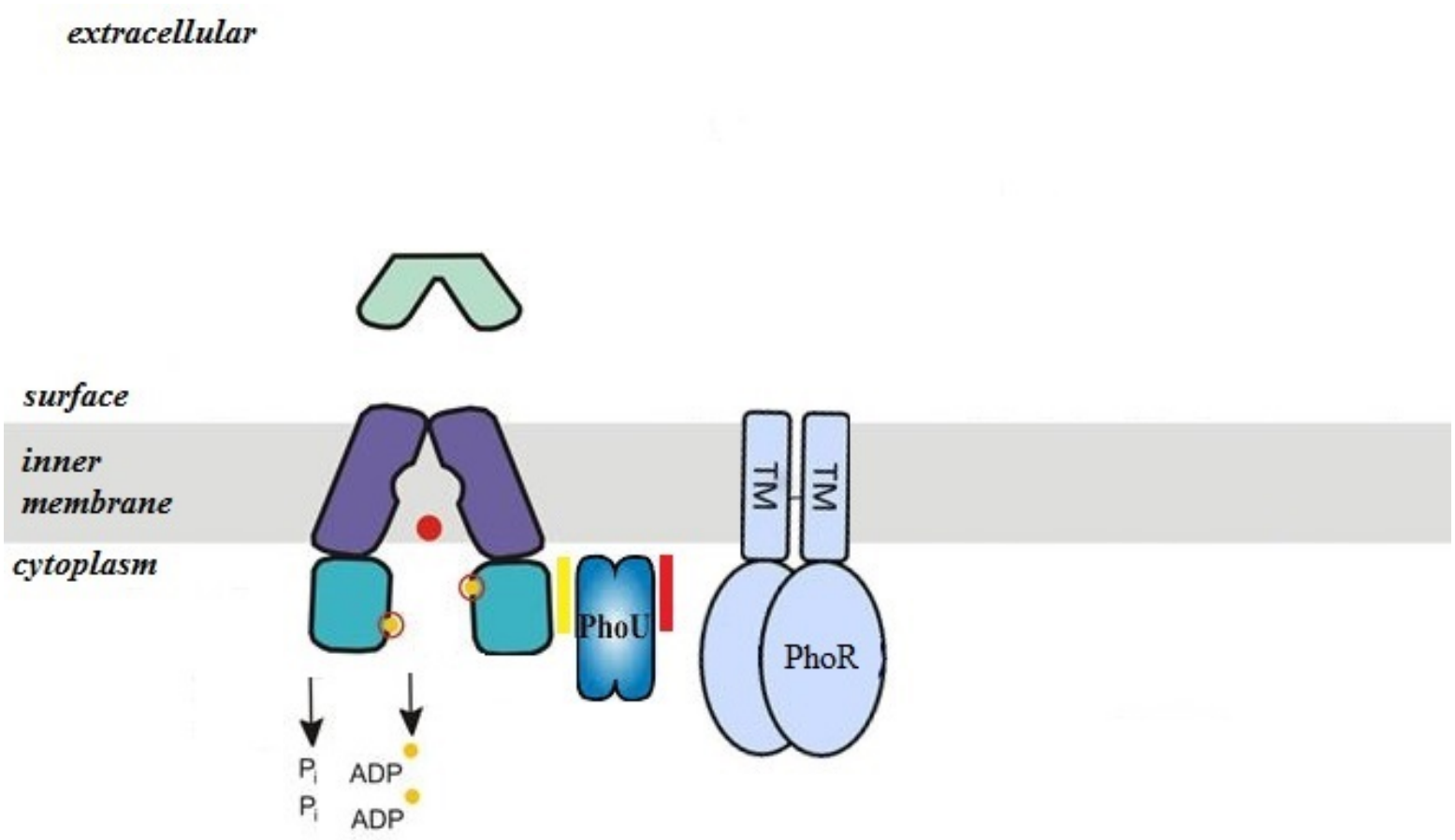

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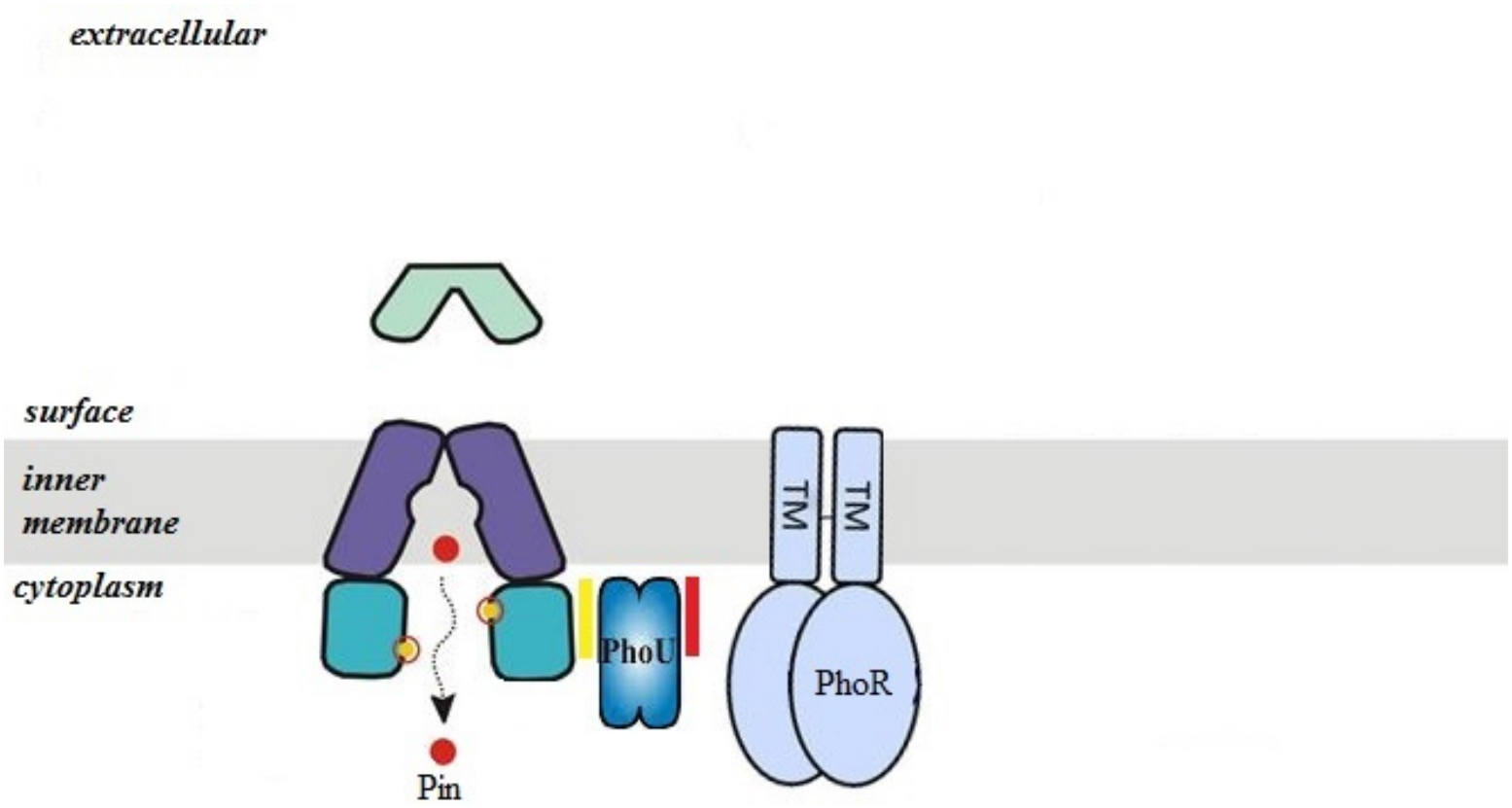

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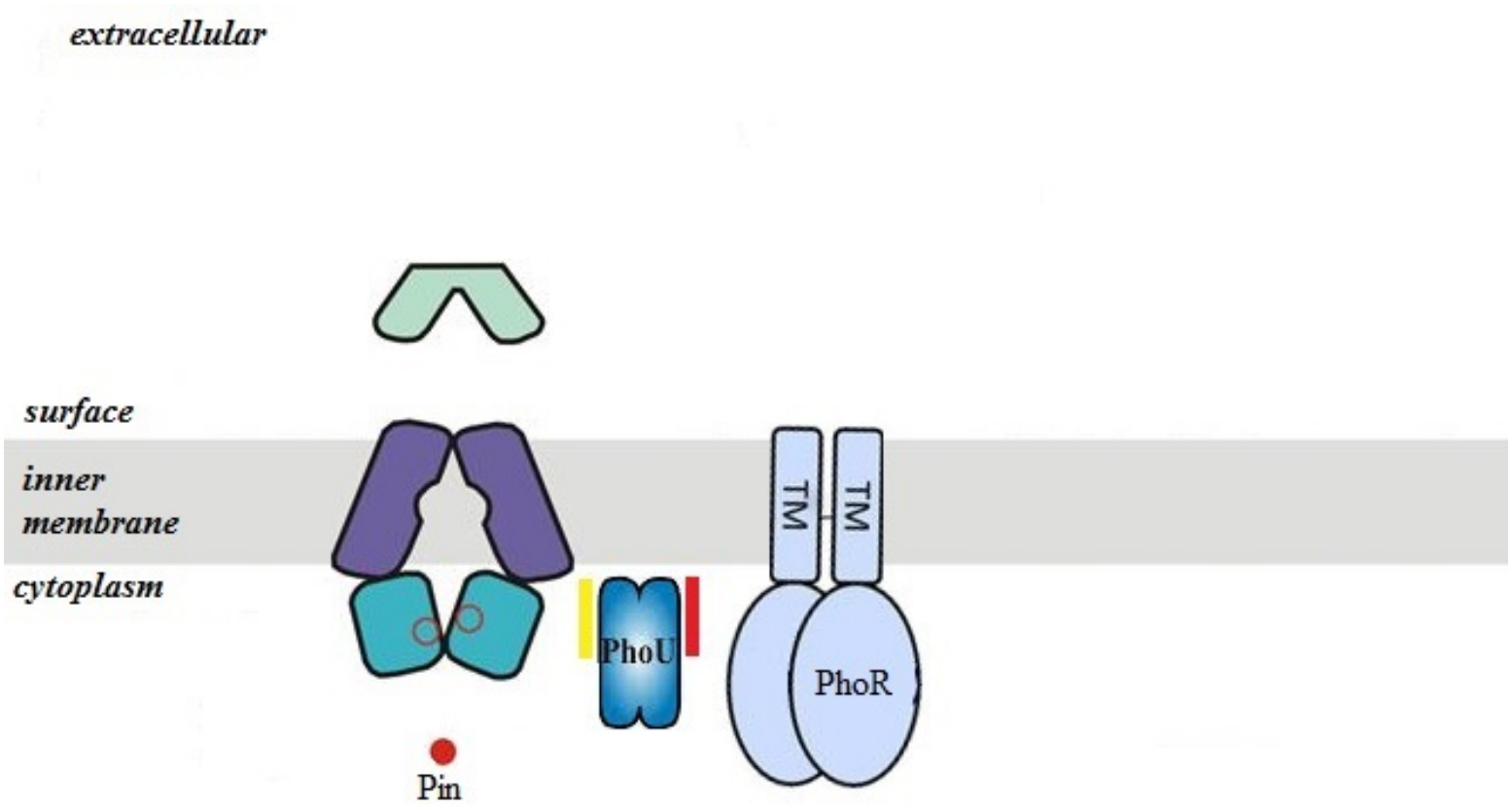

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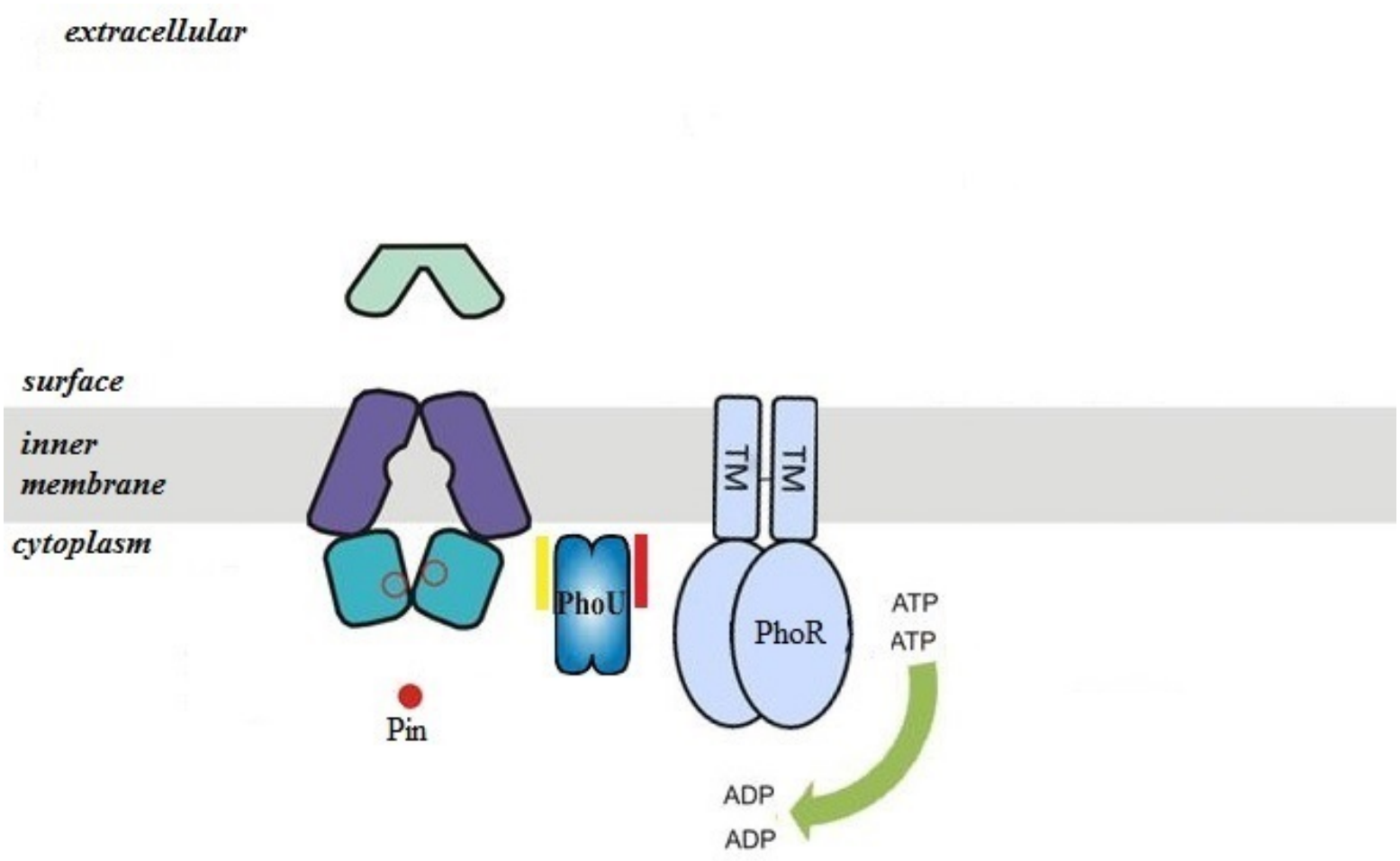

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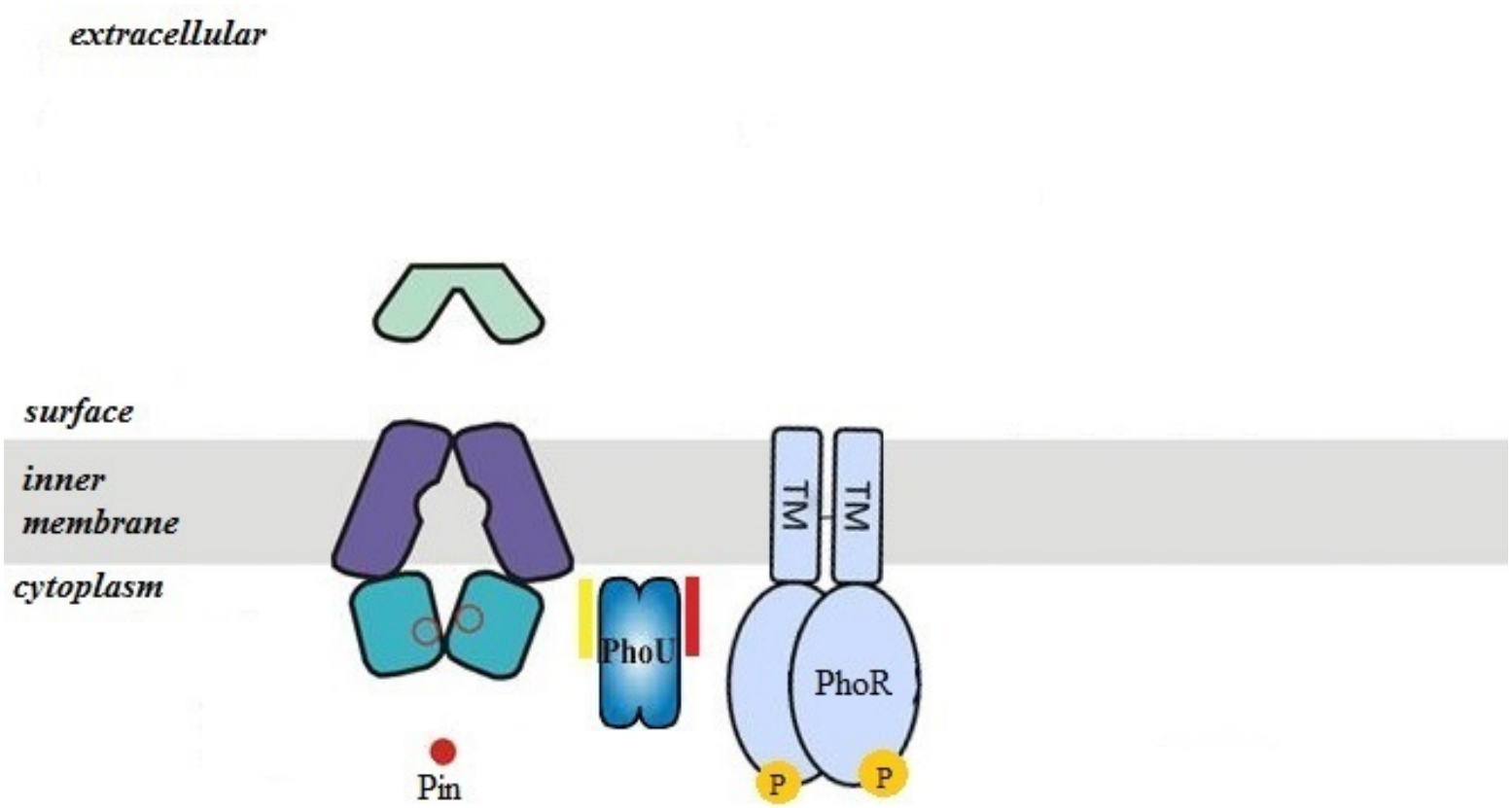

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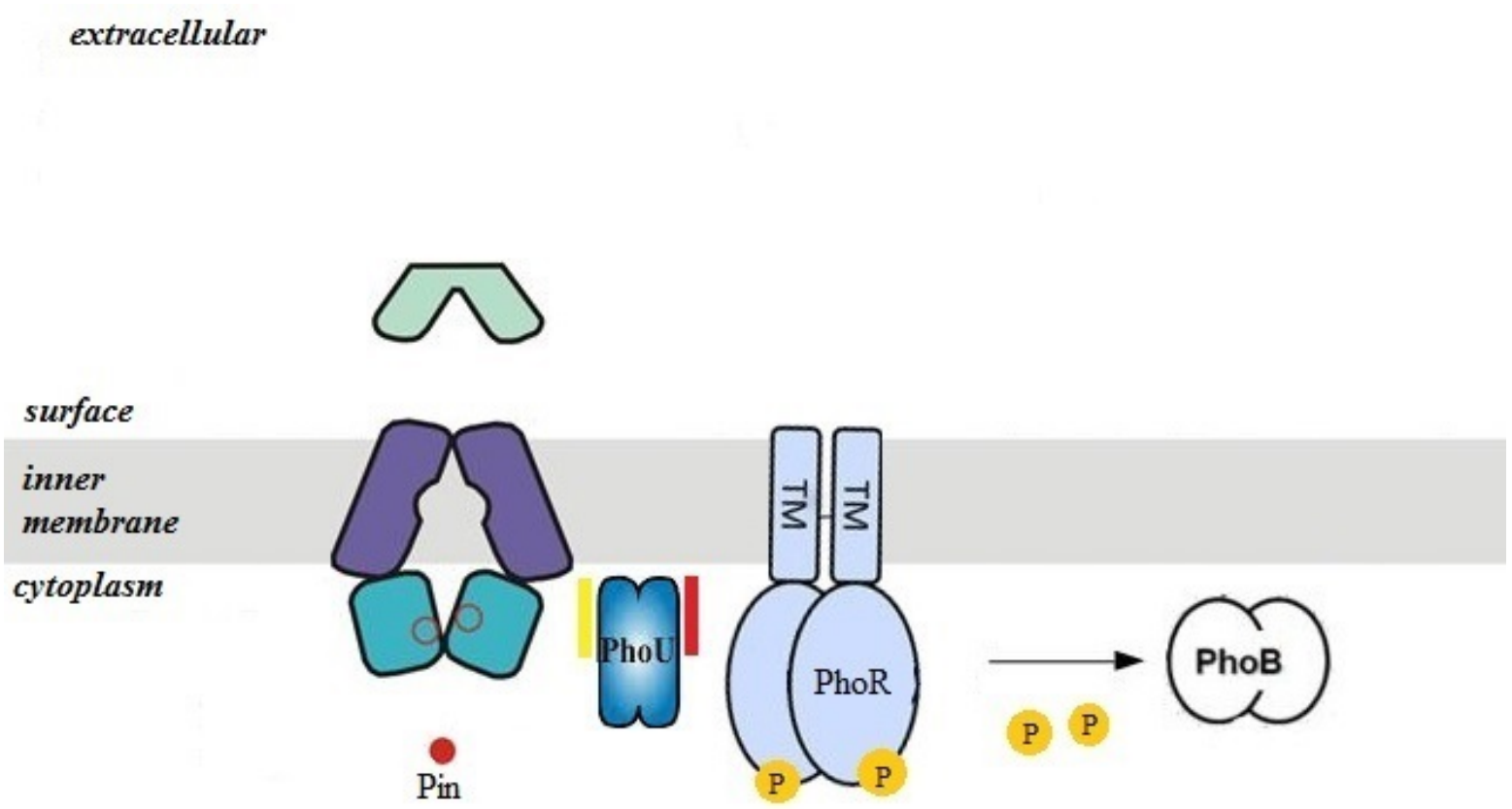

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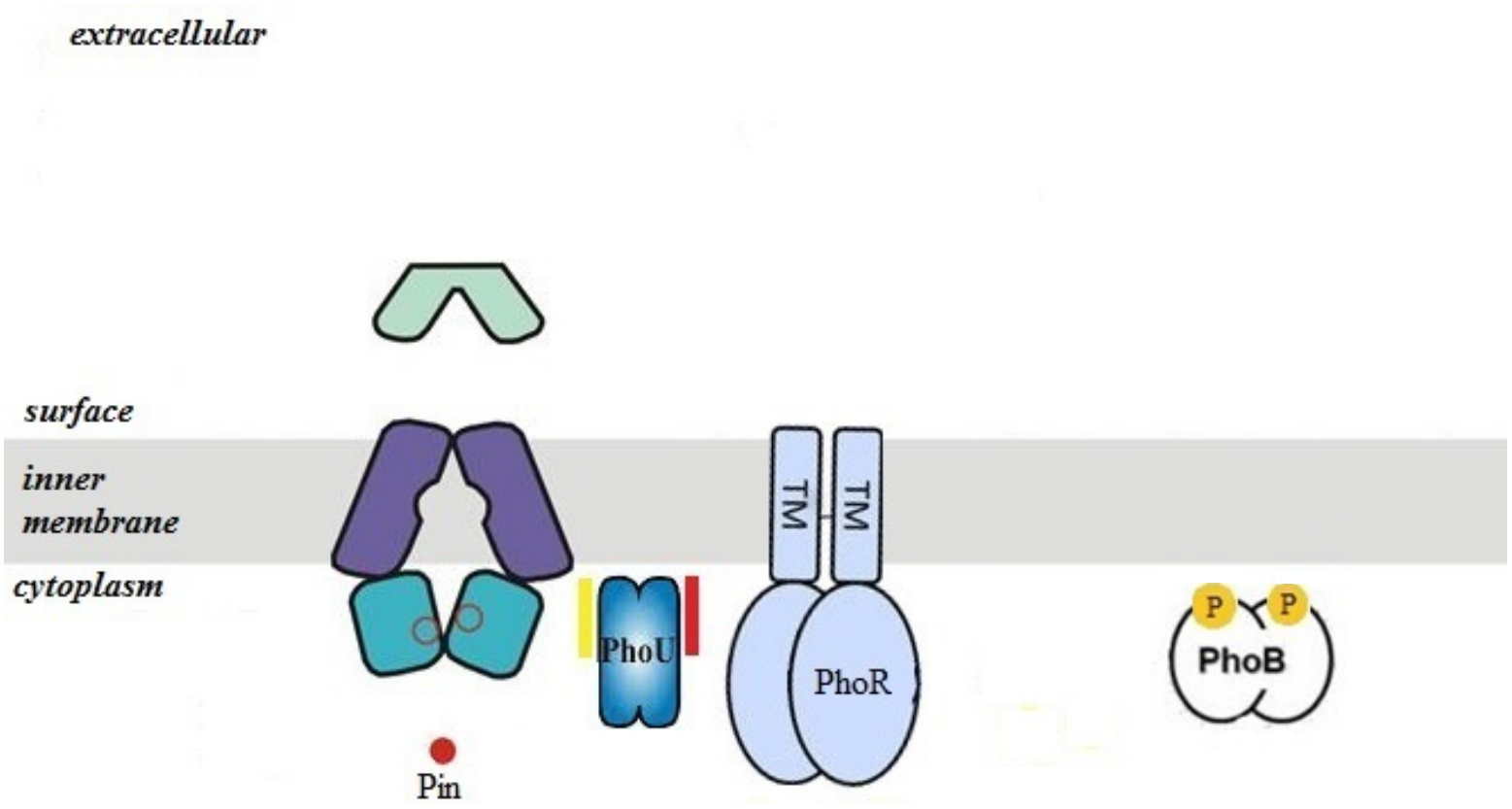

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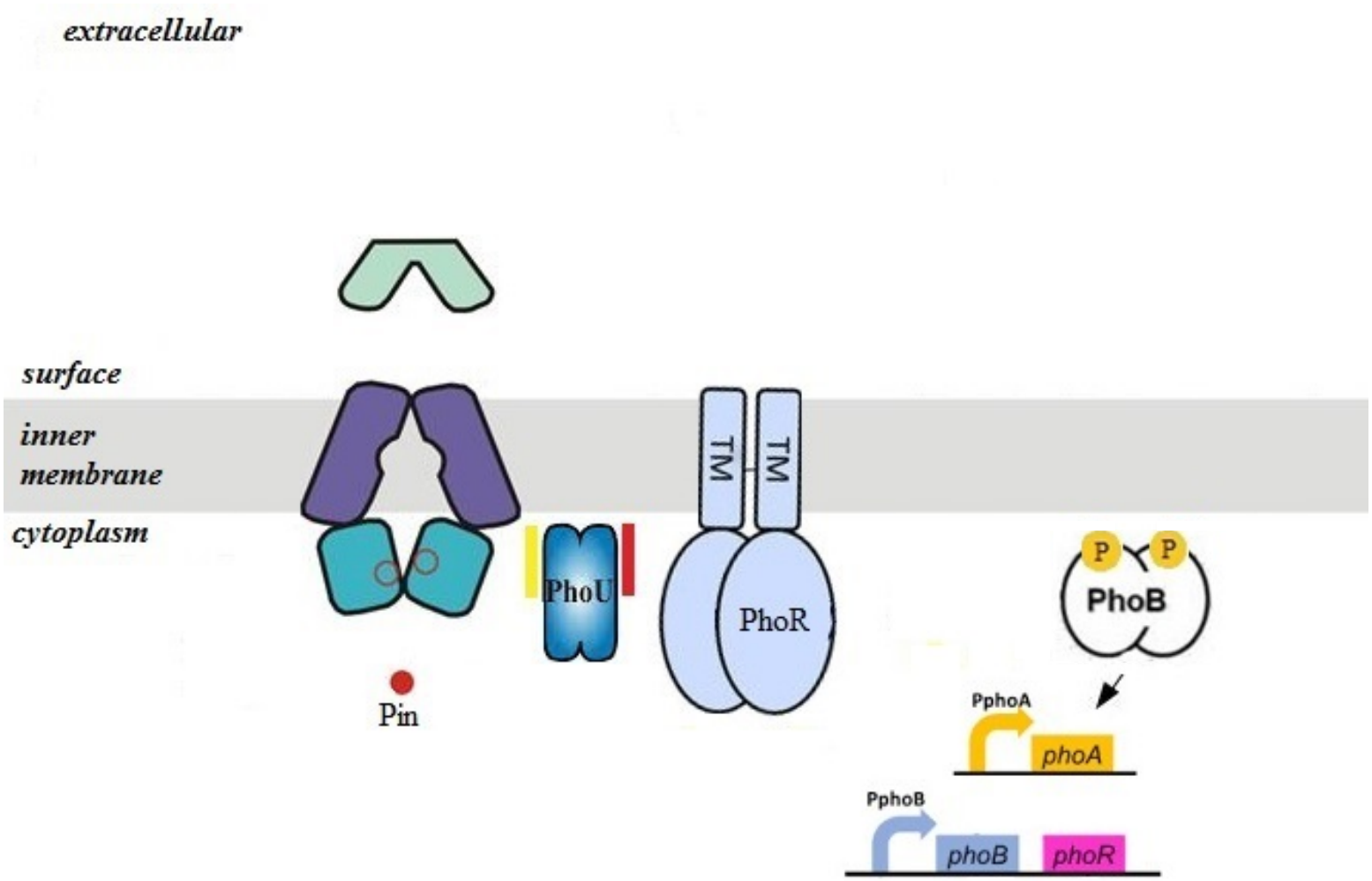
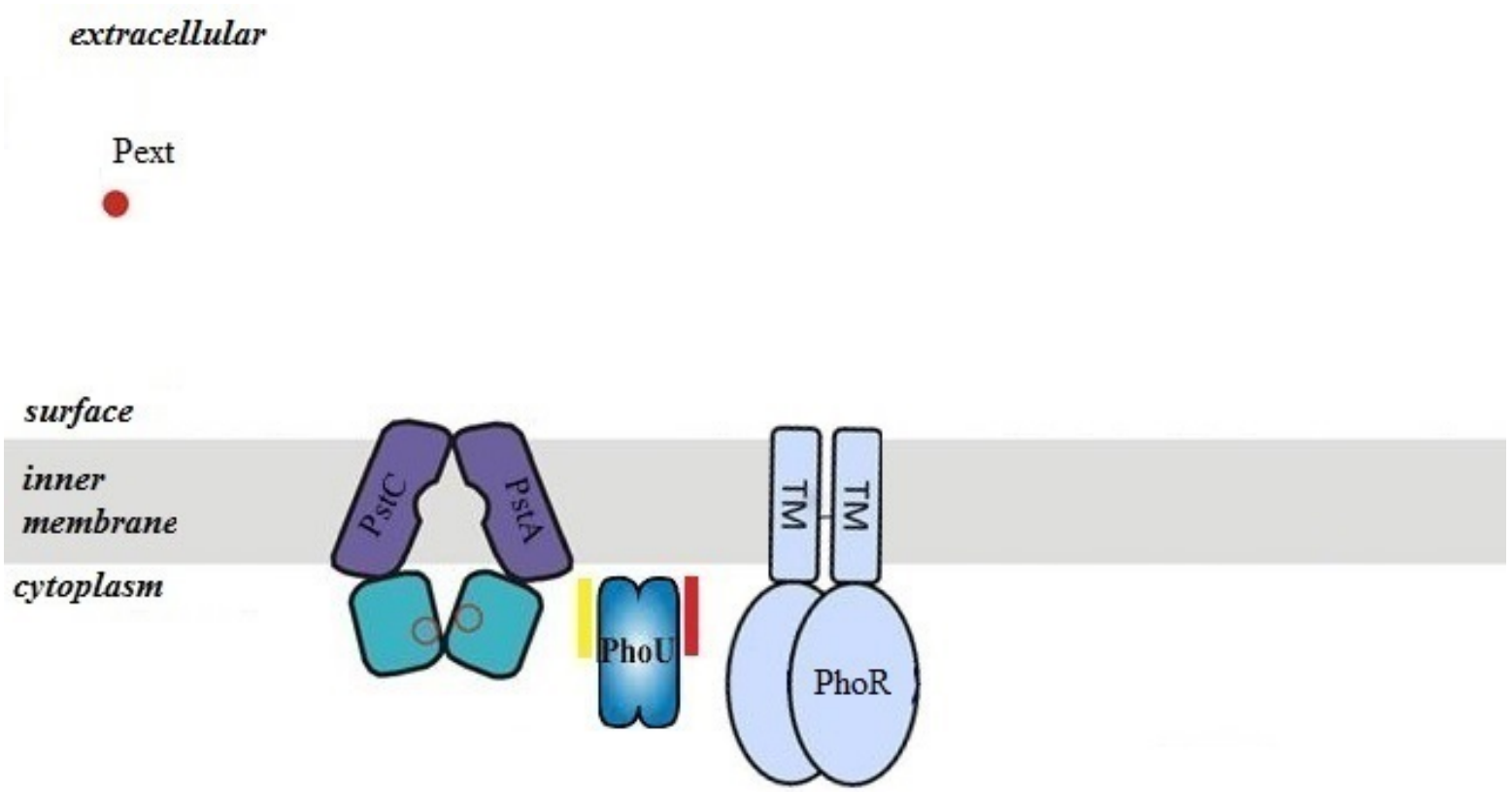


PHO Regulon when Pext is abundant

What happens when the Pext level is higher?

PHO Regulon when Pext is abundant



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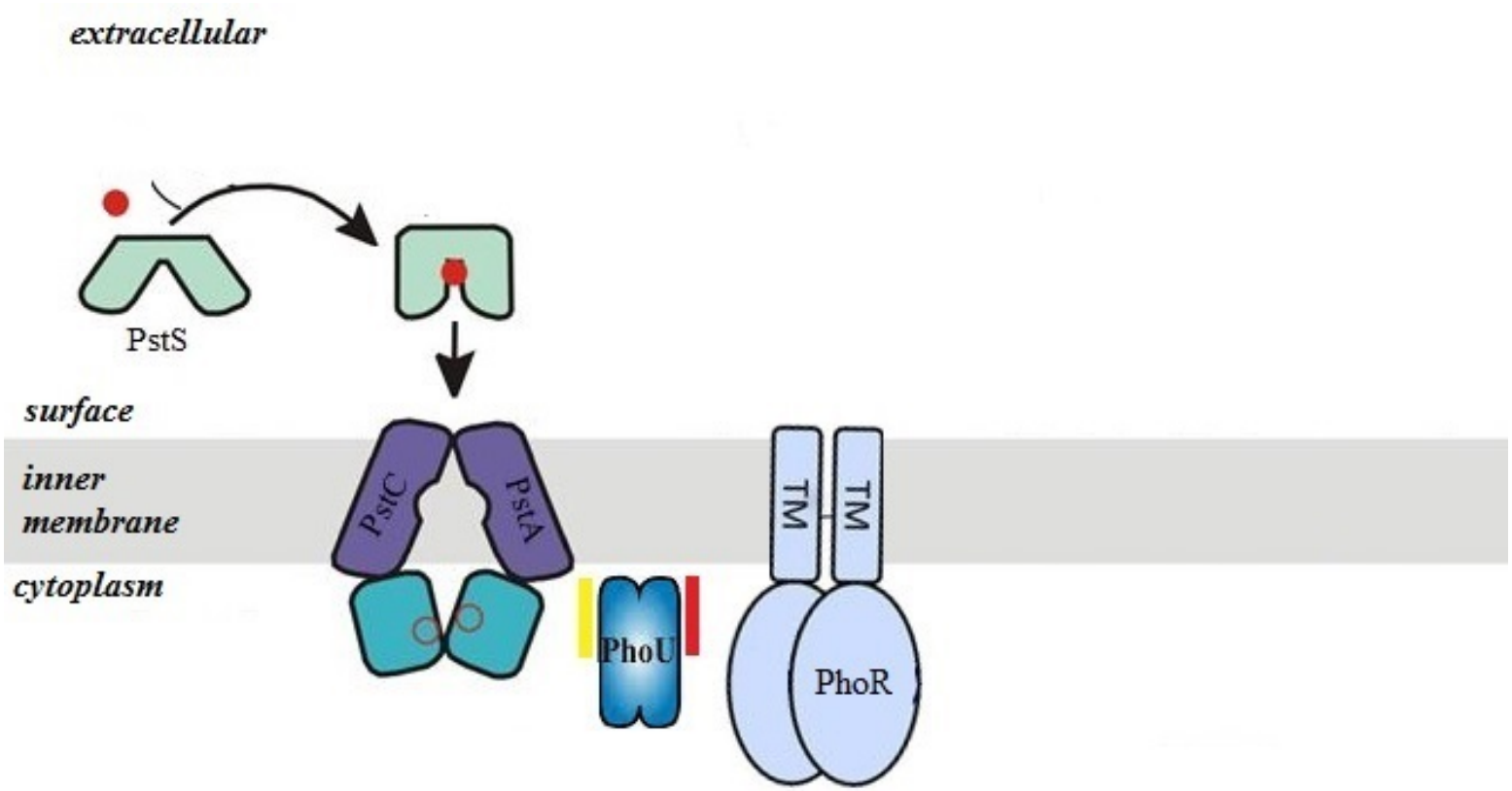

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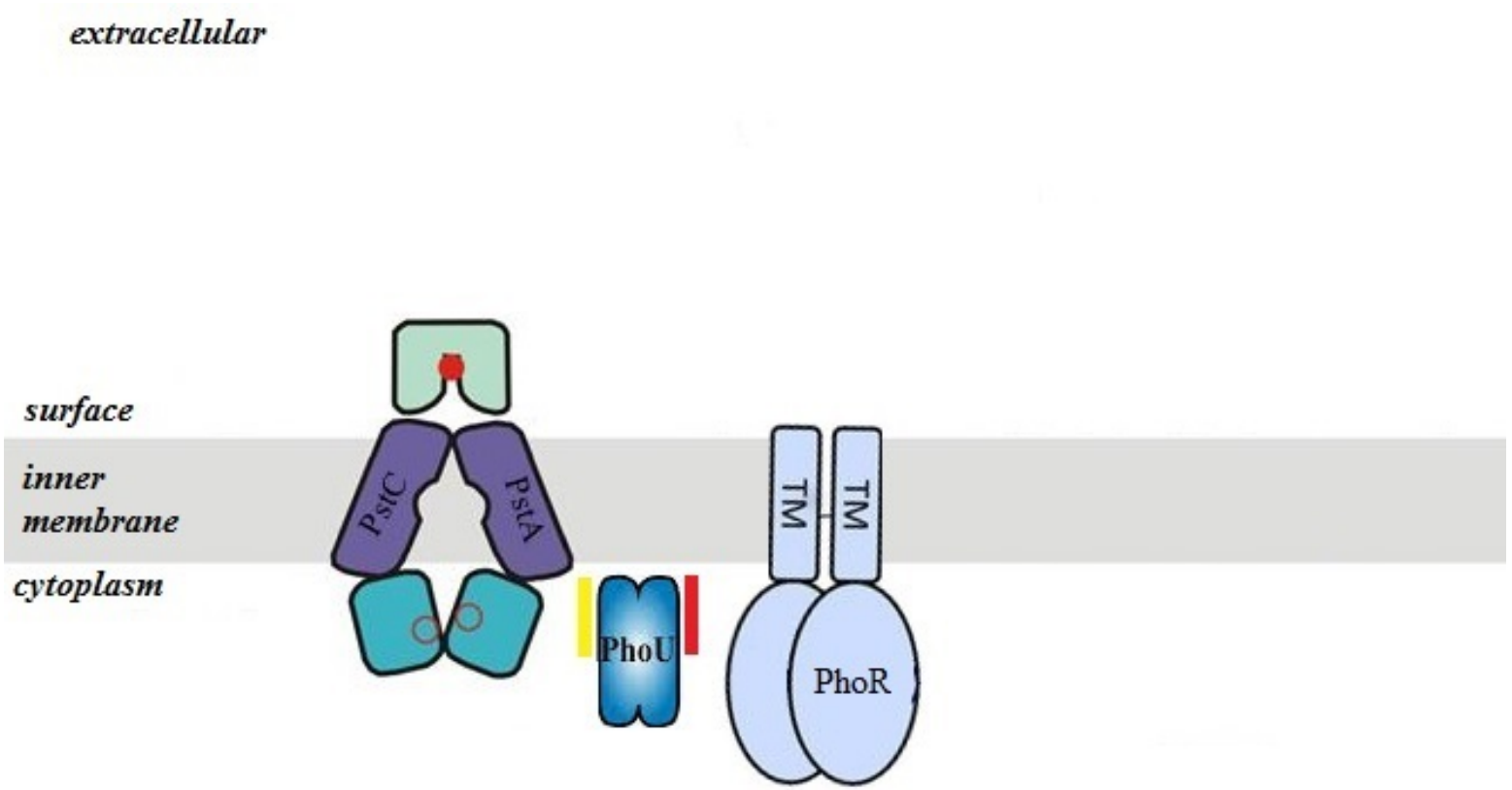

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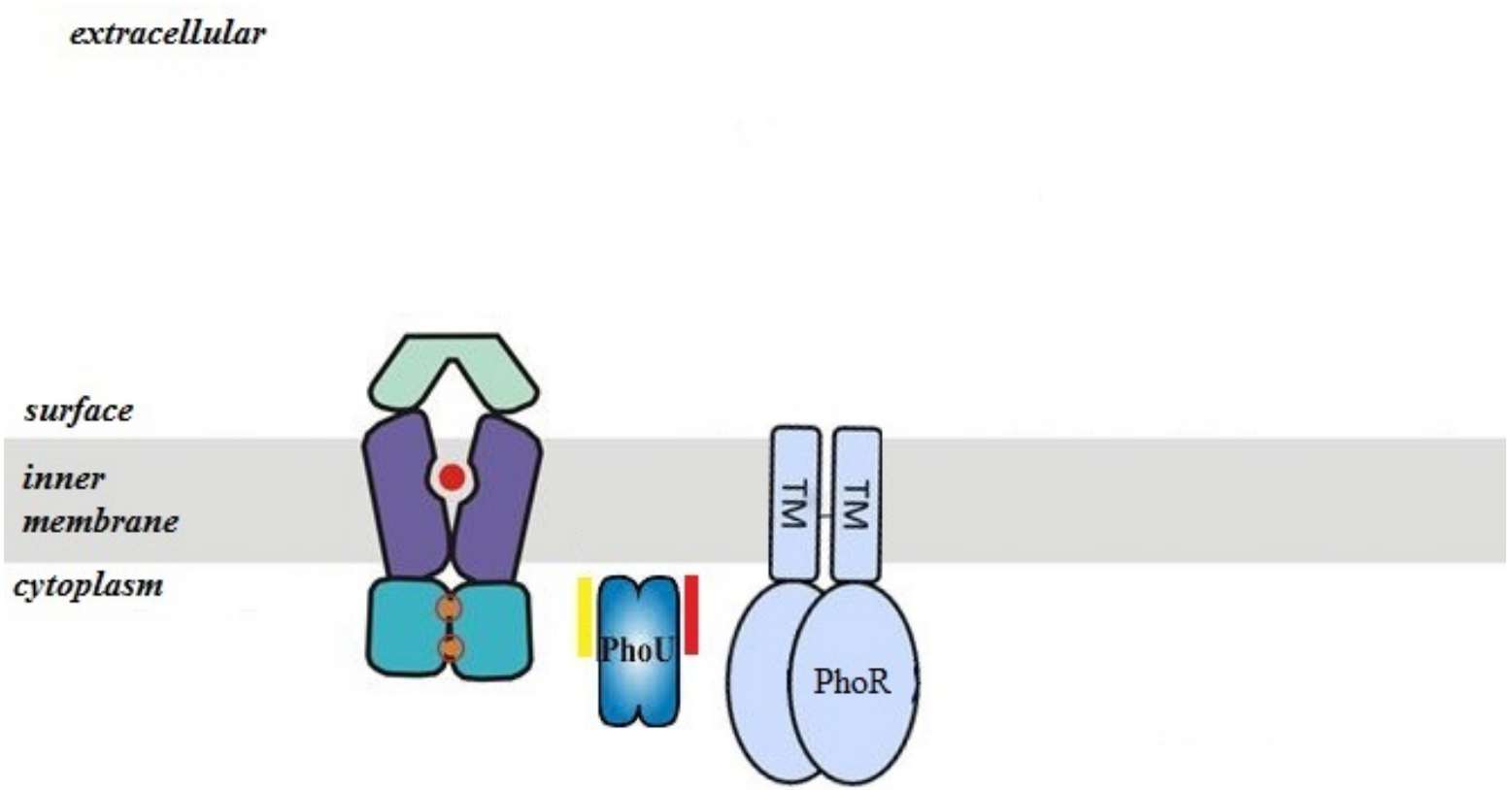

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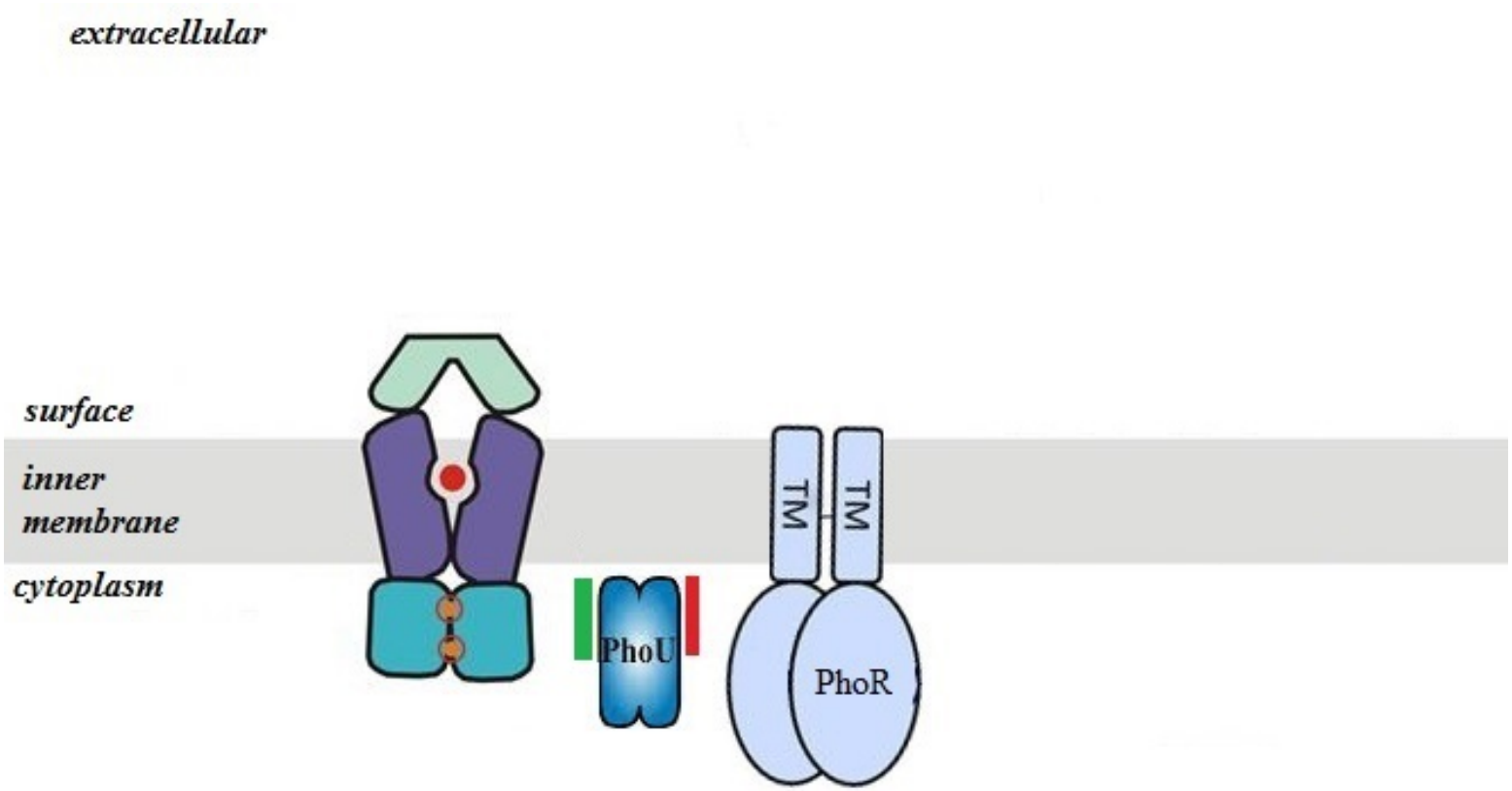

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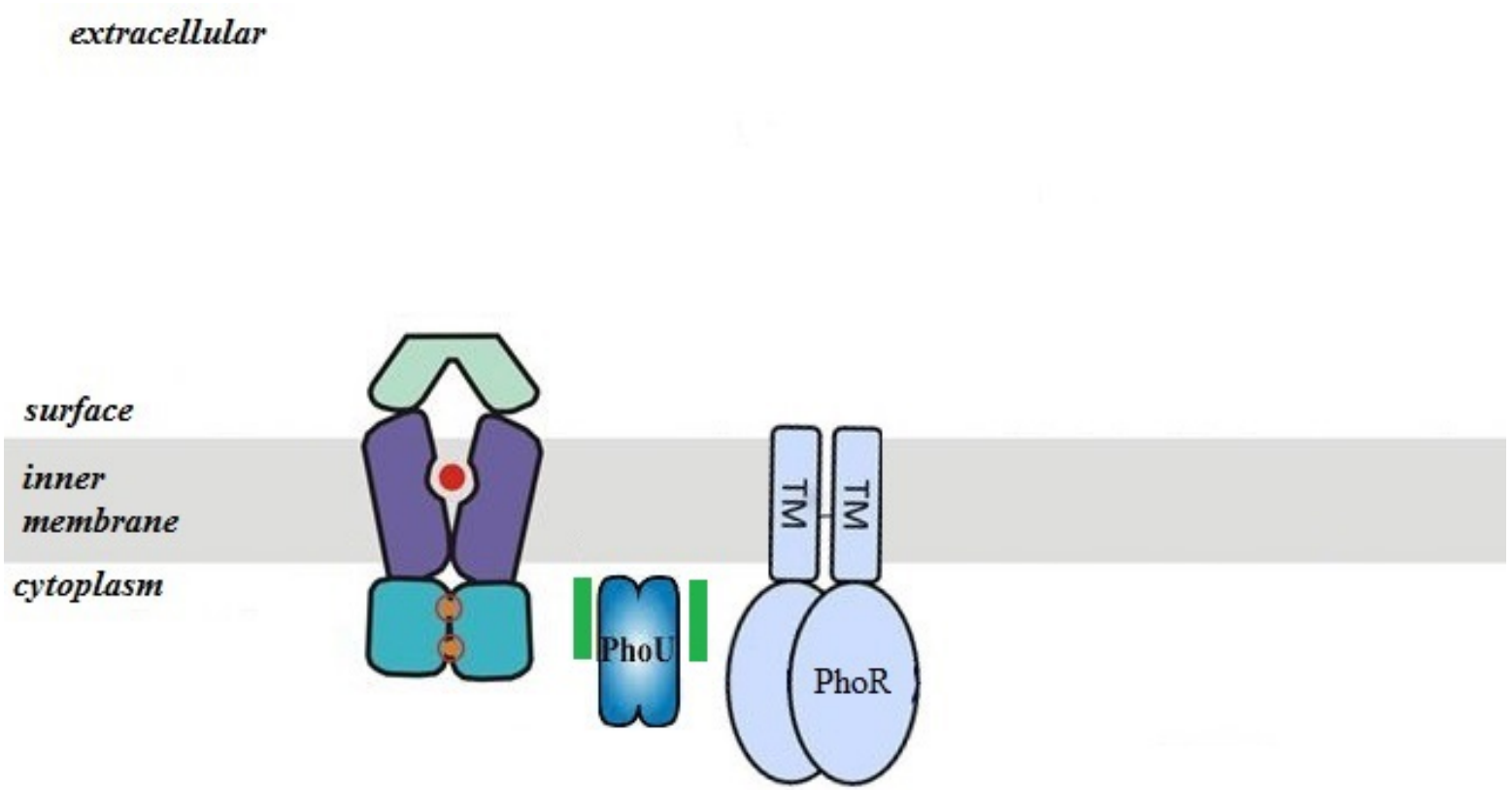

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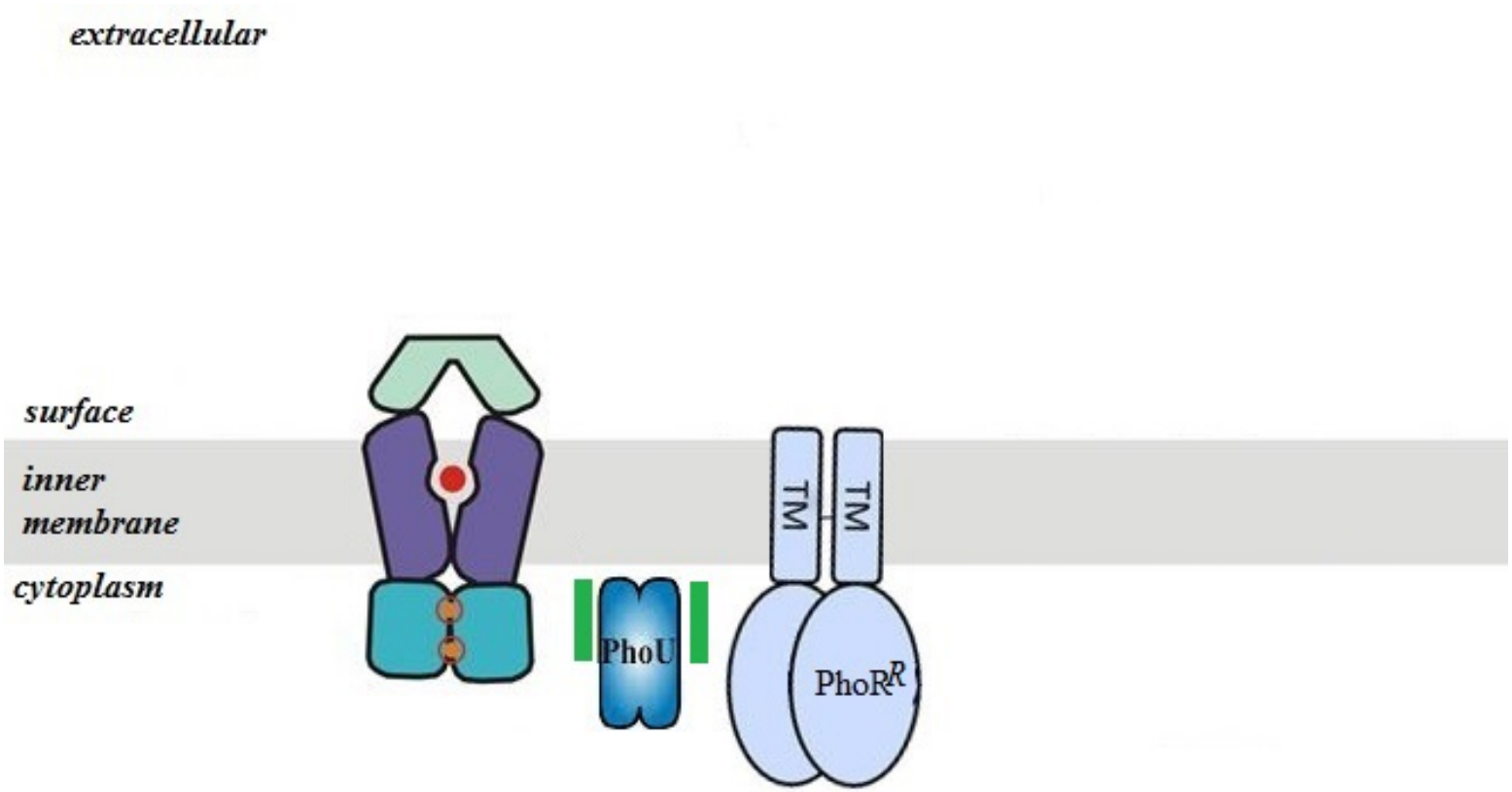

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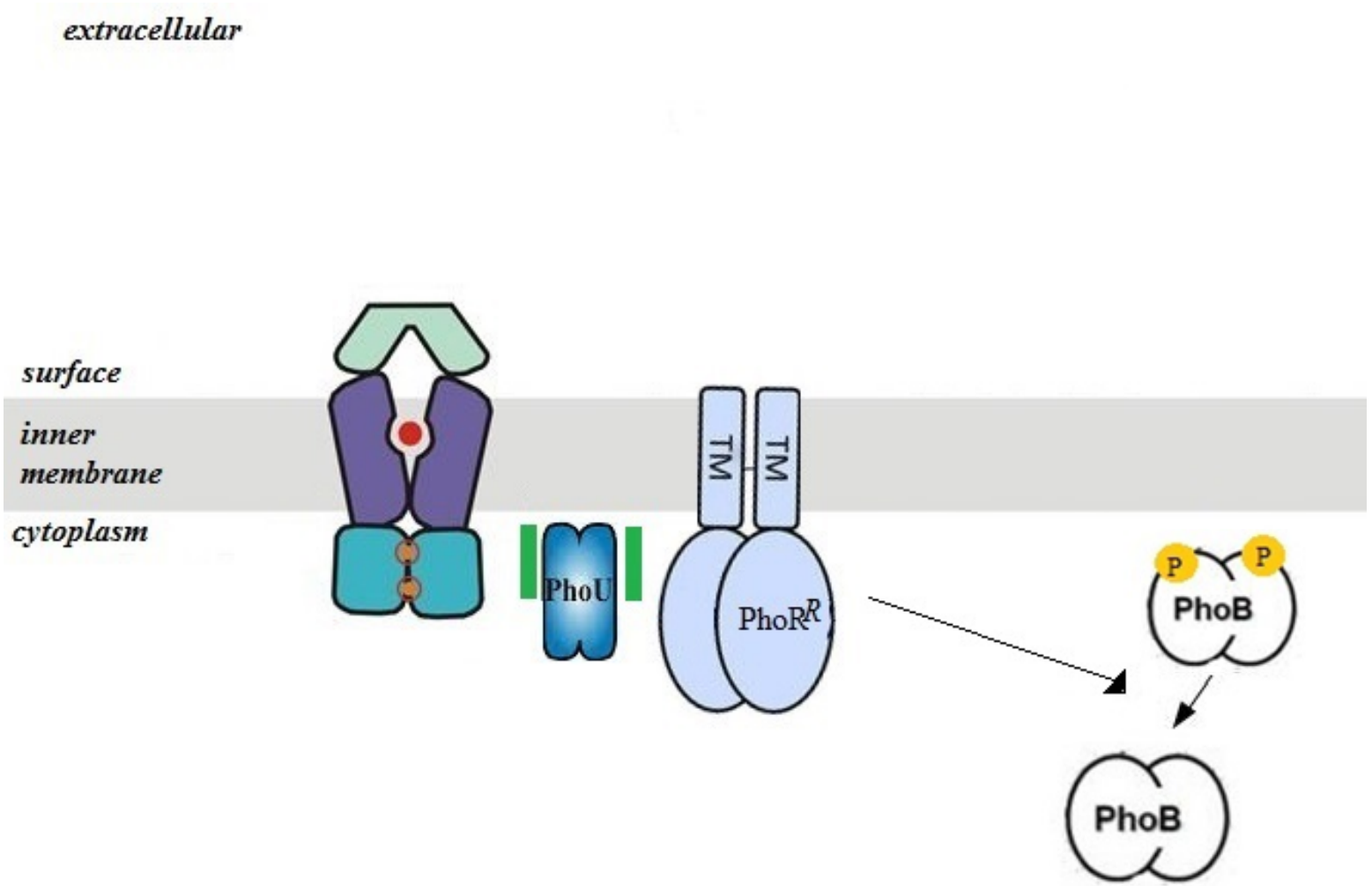
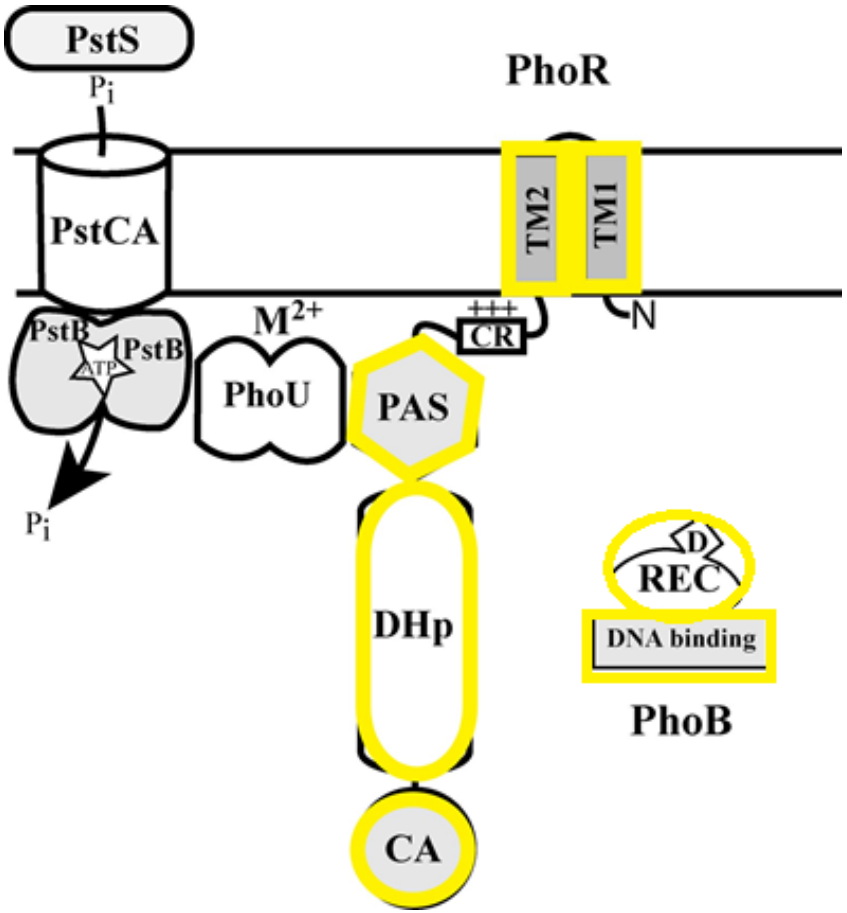
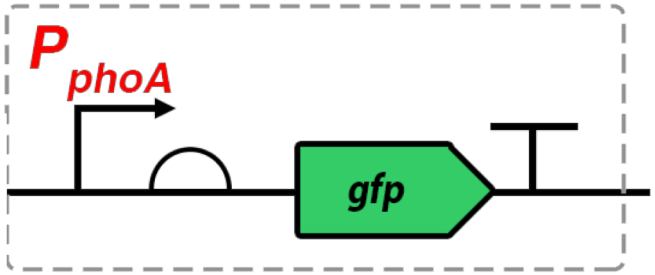
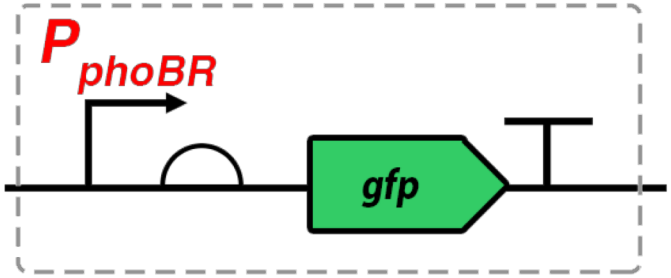


A Model of Active TCS



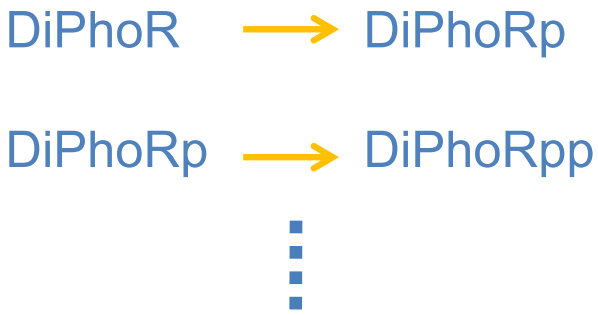
Experimental Data





Modelling of TCS response

Model ~ 29 reactions:



Modelling of TCS response

Model ~ 29 reactions:

DiPhoR → DiPhoRp

DiPhoRp → DiPhoRpp

⋮

16 ODEs

$$\begin{aligned}\mathrm{d}DiPhoR(t)/\mathrm{d}t = {} & r_{6r}.DiPhoRp(t) - r_6.DiphoR(t) \\ & + r_{11}.DiPhoRp\text{-}PhoB(t) + r_{21}.mRNAb(t) \\ & - r_{24}.DiPhoR(t) - r_{13}.DiPhoR(t).PhoBp(t) \\ & + r_{13r}.DiPhoR\text{-}PhoBp(t) + r_{14}.DiPhoR\text{-}PhoBp(t)\end{aligned}$$

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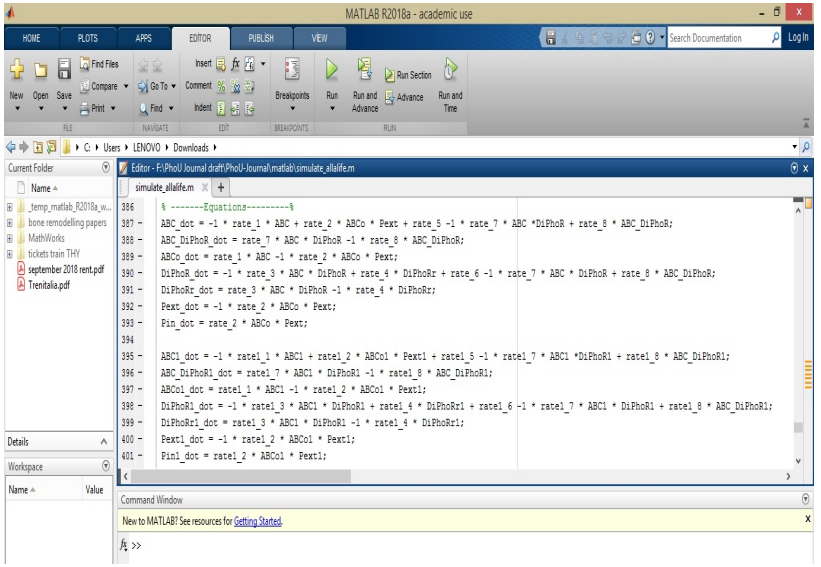
Matlab code



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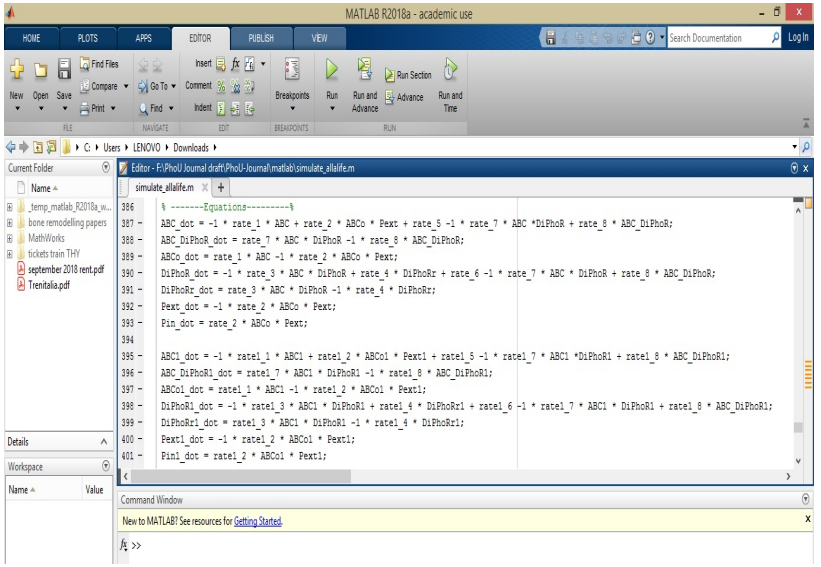
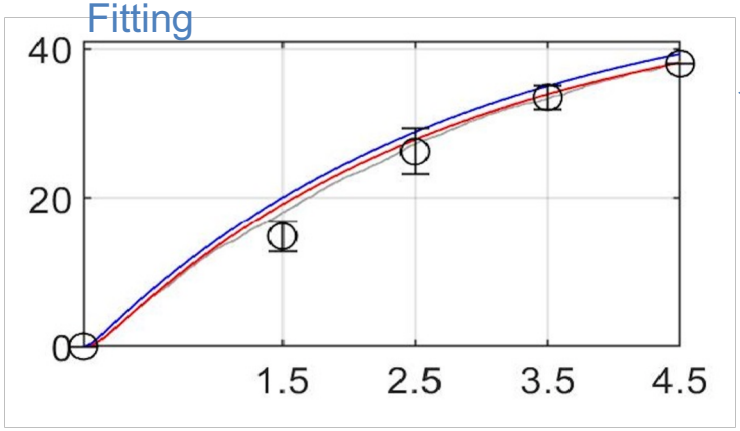
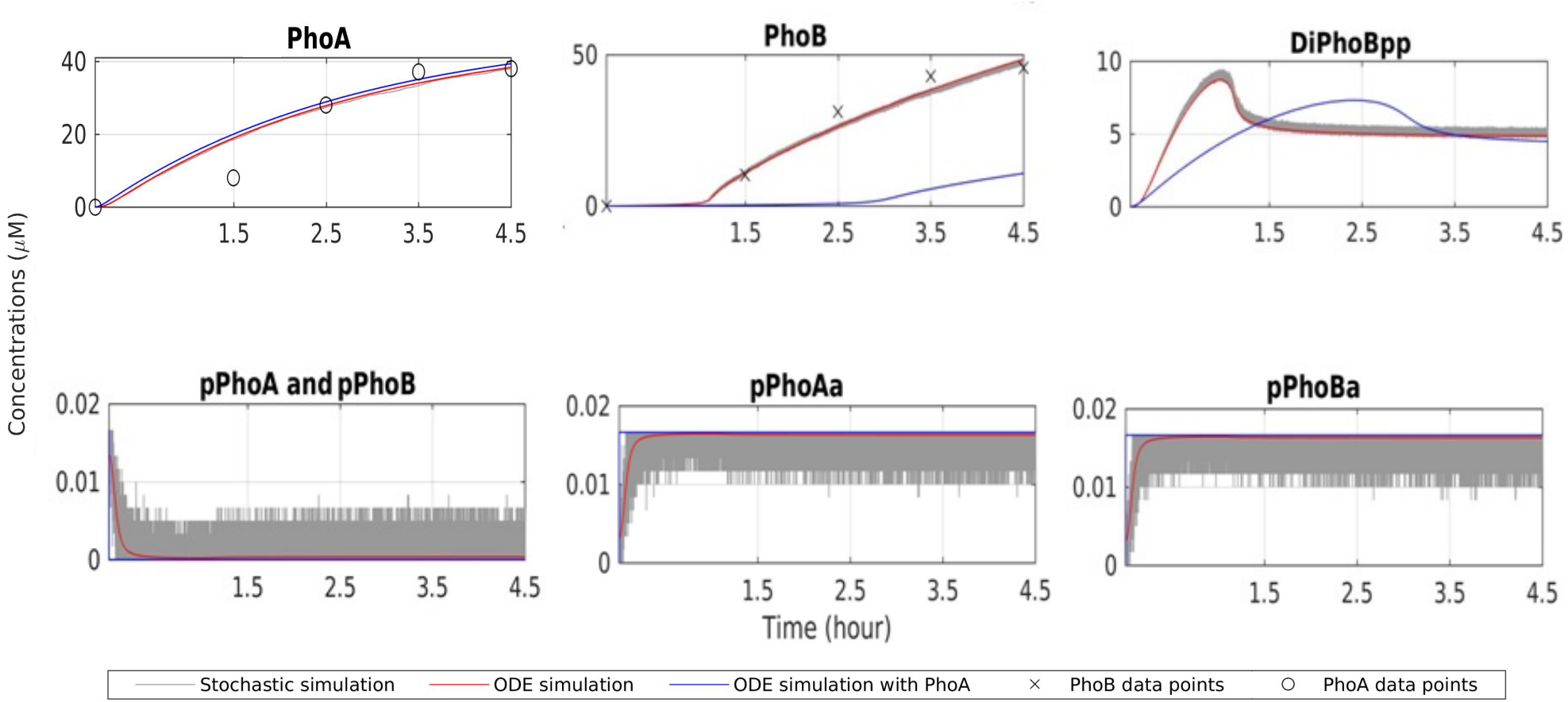


Fitting



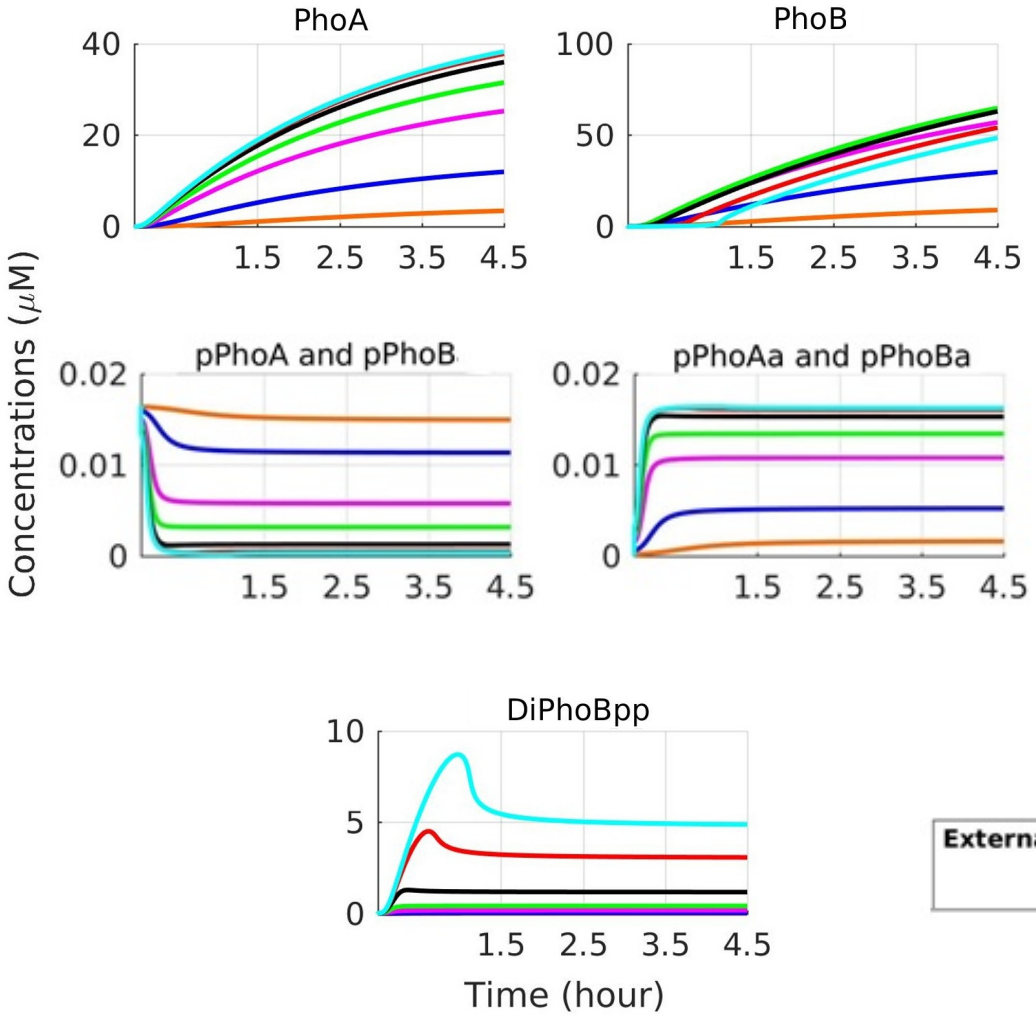
Simulation of Active TCS

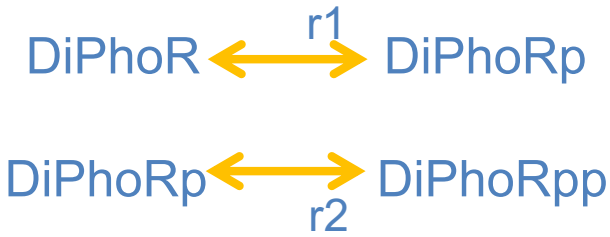


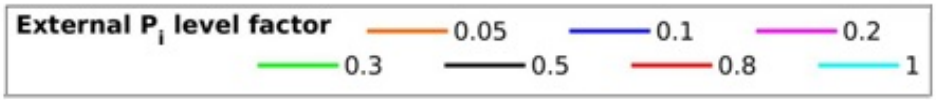
Uluşeker et al., *A Dynamic Model of the Phosphate Response System with Synthetic Promoters in E. coli*. **ECAL** 2017

Uluşeker et al. Dynamic mechanisms of phosphate response with synthetic promoters in *Escherichia coli*, ***Scientific Reports***, *2019, in press*

Simulation of the System







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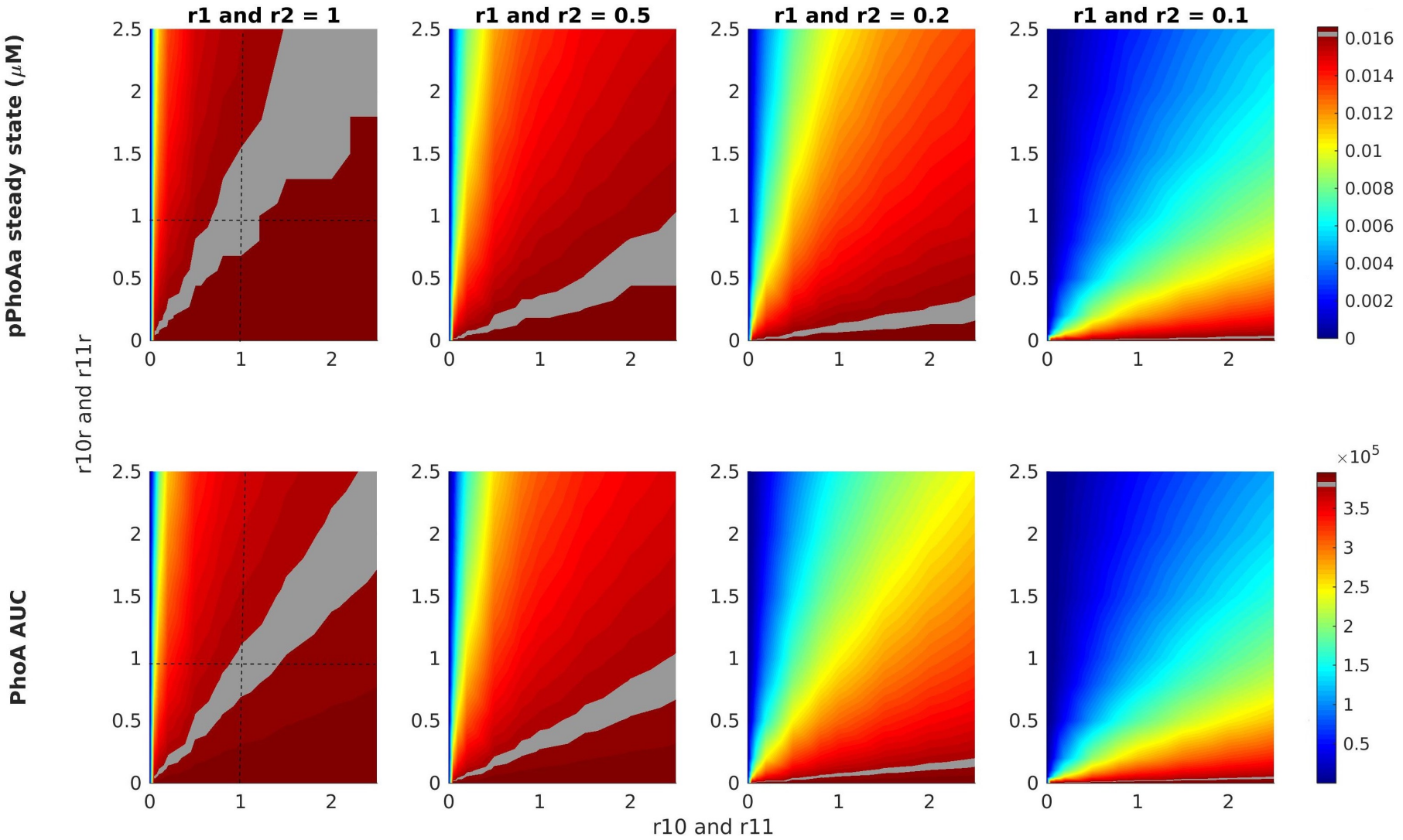
Simulation with Different Pext Levels



DiPhoBpp +pPhoA r10r ⟷ r10 pPhoAa

DiPhoBpp +pPhoB r11r ⟷ r11 pPhoBa

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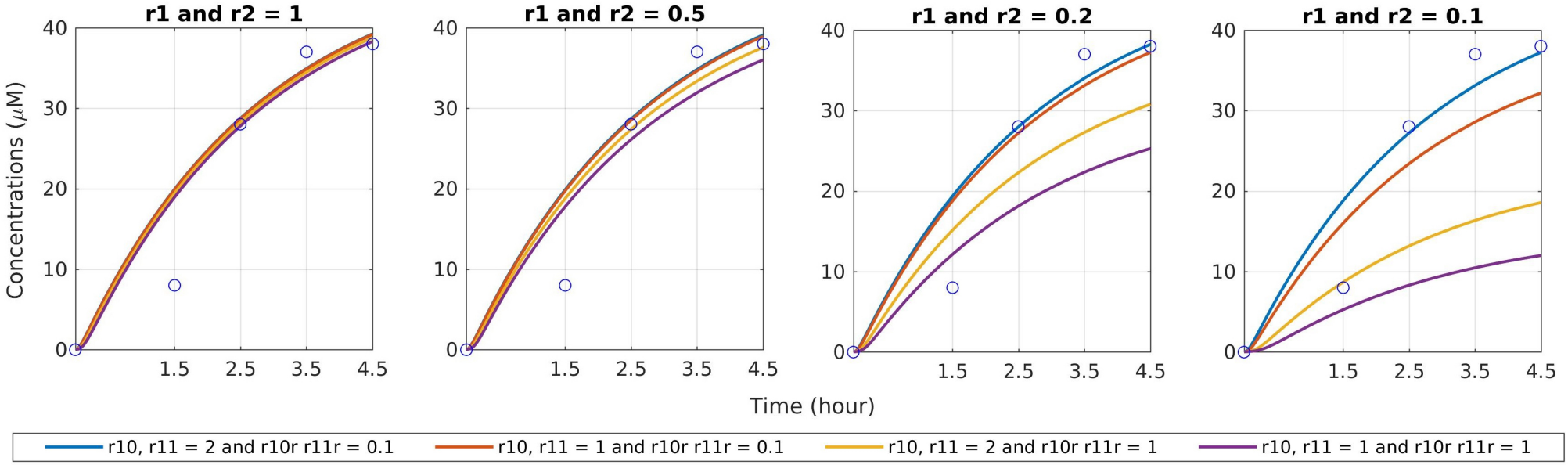
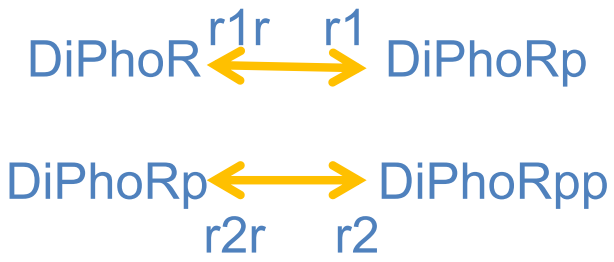
Simulation with Different Pext Levels





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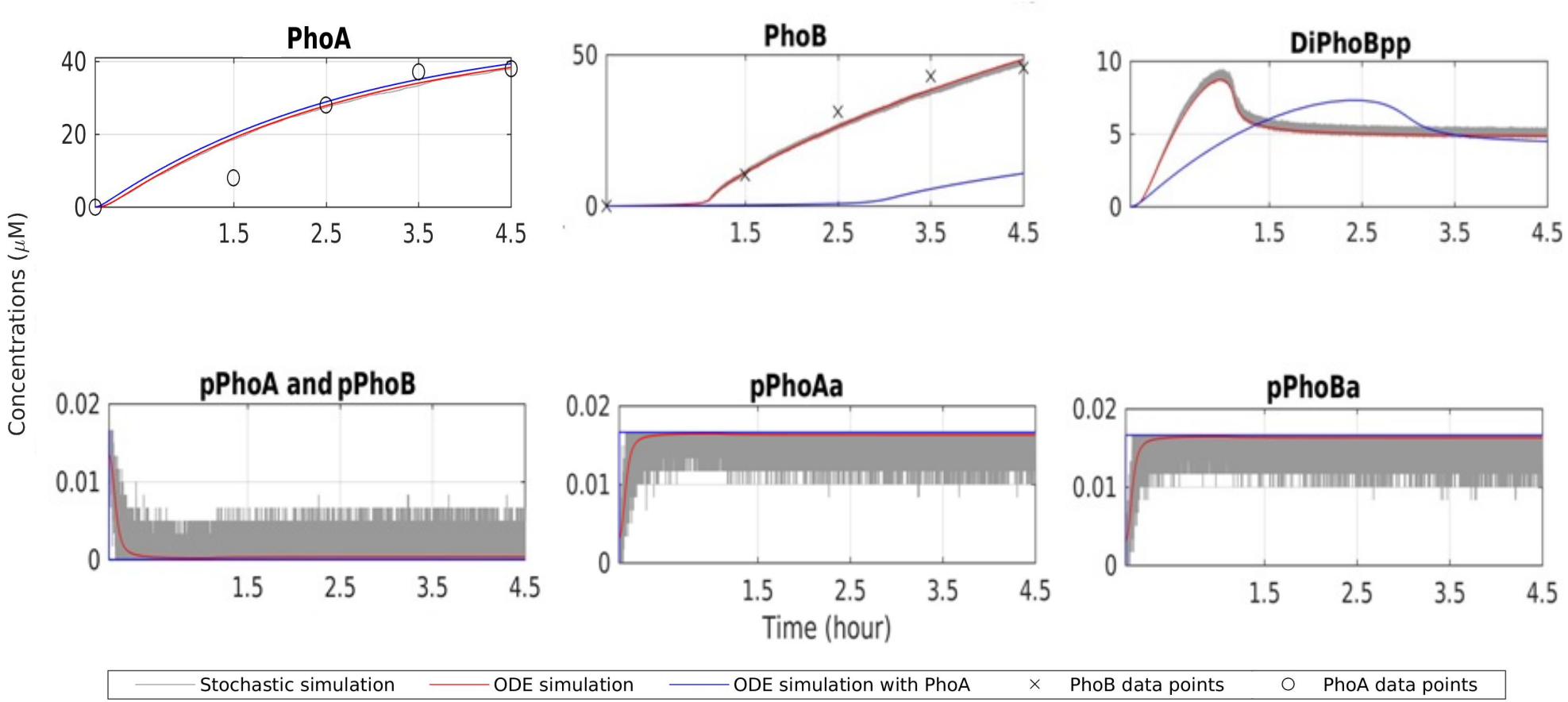

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Stochastic Flux Analysis

A simulation is a reduction of a complex structure.

Flux is the flow of resources in a simulation.

Ozan Kahramanoğulları, Quantifying Information Flow in Chemical Reaction Networks, **AICoB**, 2017.

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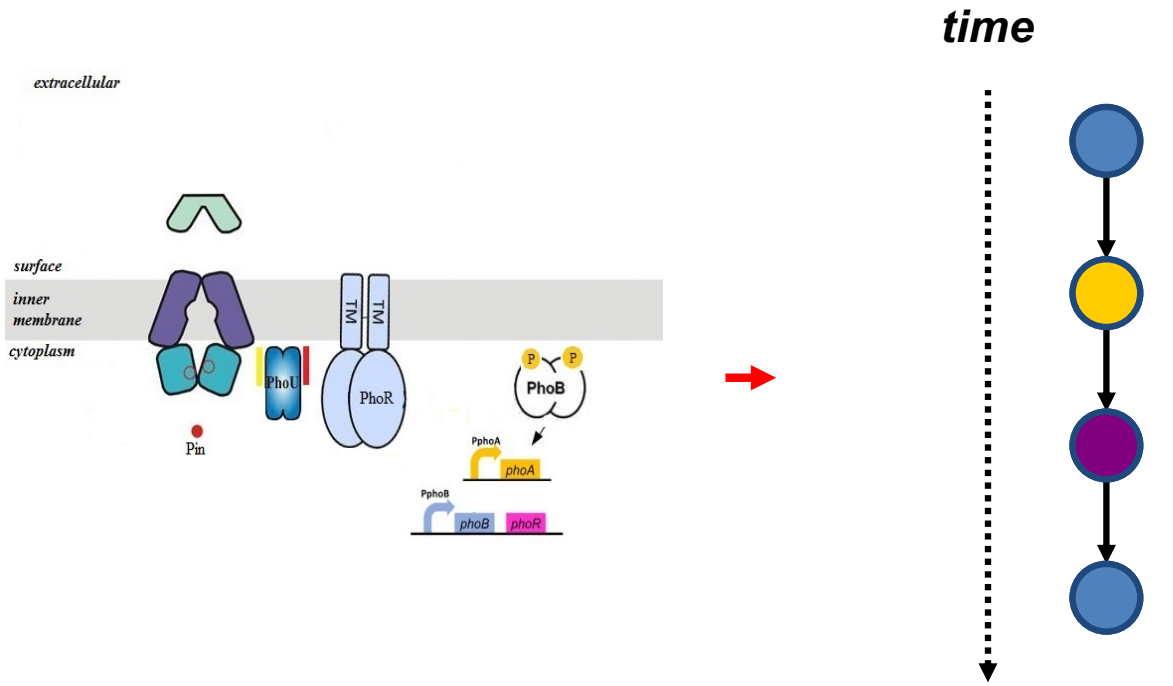

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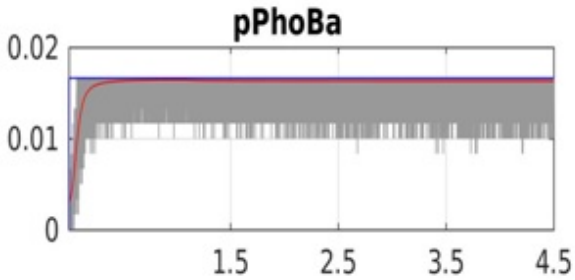
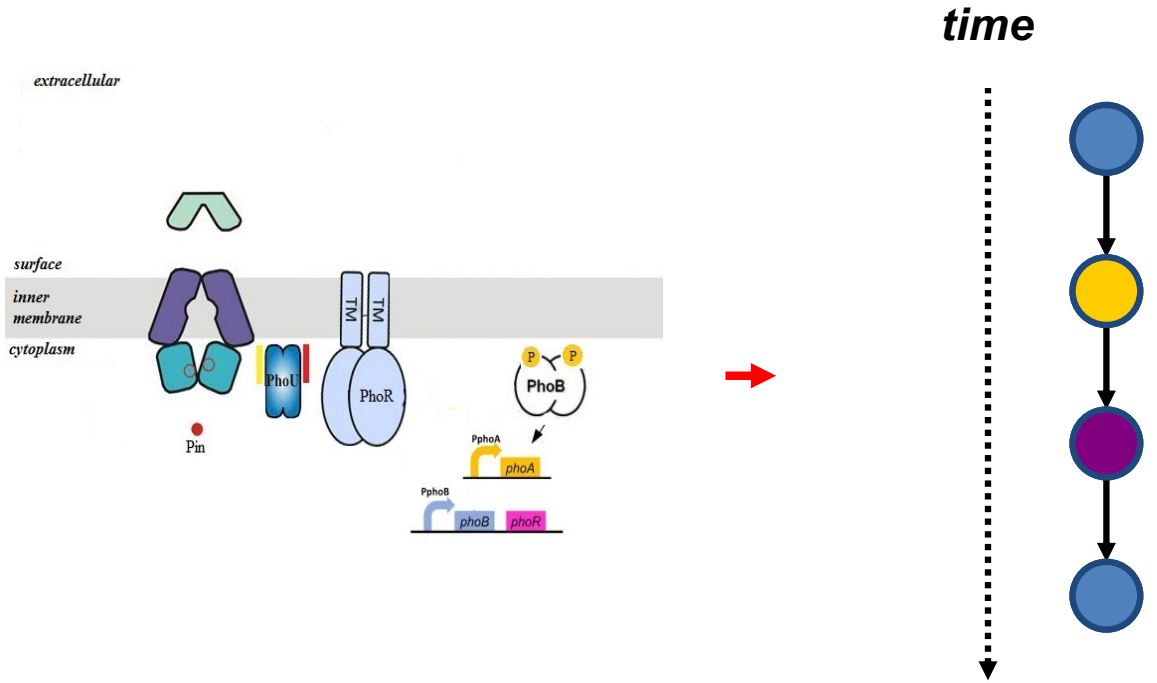




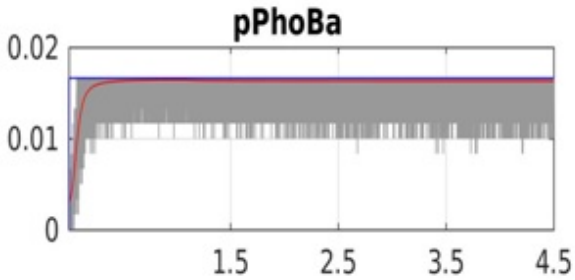
Ozan Kahramanoğulları, Quantifying Information Flow in Chemical Reaction Networks, **AICoB**, 2017.

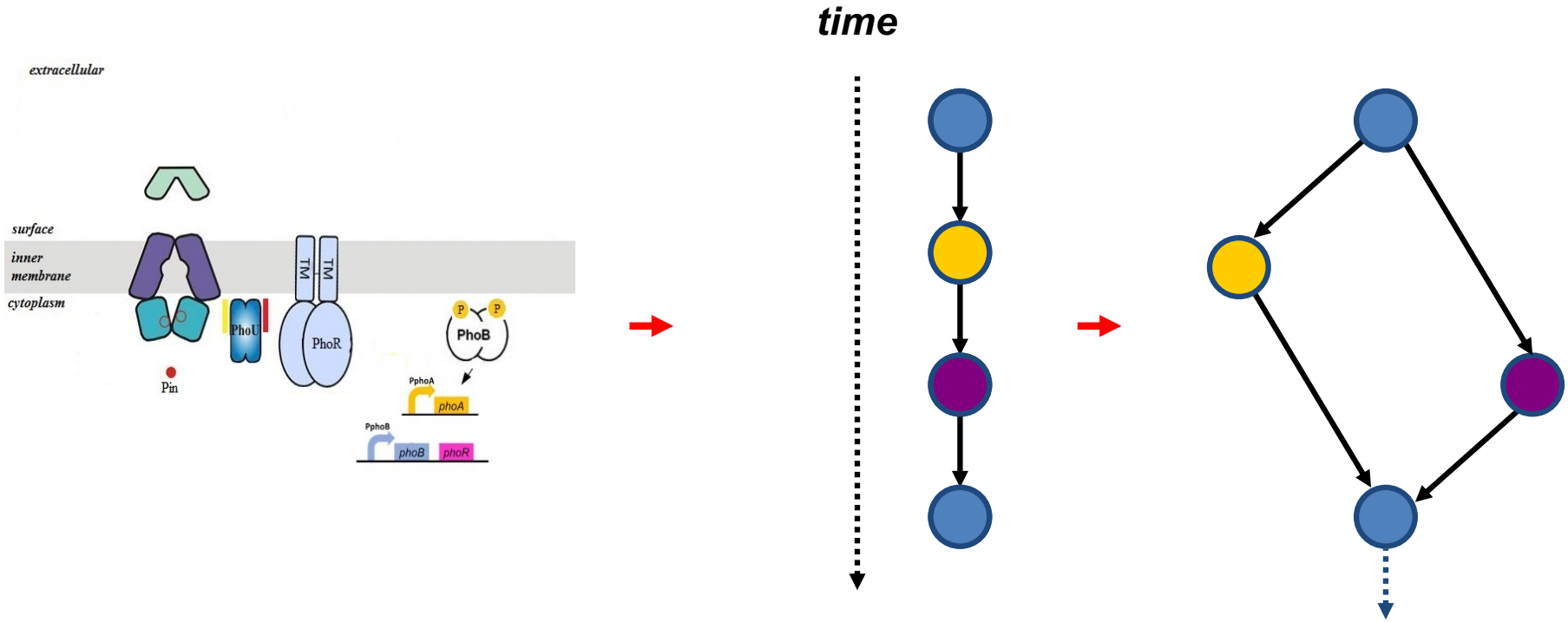
Ozan Kahramanoğulları, James Lynch, Stochastic Flux Analysis of Chemical Reaction Networks, **BMC Systems Biology**, 2013.

Stochastic Flux Analysis

A simulation is a reduction of a complex structure.

Flux is the flow of resources in a simulation.





Ozan Kahramanoğulları, Quantifying Information Flow in Chemical Reaction Networks, **AICoB**, 2017.

Ozan Kahramanoğulları, James Lynch, Stochastic Flux Analysis of Chemical Reaction Networks, **BMC Systems Biology**, 2013.

Stochastic Flux Analysis – first 15 minutes

1 DiPhoR → DiPhoRp
2 DiPhoRp → DiPhoR
3 DiPhoRp → DiPhoRpp
4 DiPhoRpp → DiPhoRp
5 DiPhoRpp + PhoB → DiPhoRpp_PhoB
6 DiPhoRpp_PhoB → DiPhoRpp + PhoB
7 DiPhoRpp_PhoB → DiPhoRp + PhoBp
8 DiPhoRp + PhoB → DiPhoRp_PhoB
9 DiPhoRp_PhoB → DiPhoRp + PhoB
10 DiPhoRp_PhoB → DiPhoR + PhoBp
11 PhoBp + PhoBp → DiPhoBpp
12 DiPhoBpp → PhoBp + PhoBp
13 DiPhoR + PhoBp → DiPhoR_PhoBp
14 DiPhoR_PhoBp → DiPhoR + PhoBp
15 DiPhoR_PhoBp → DiPhoR + PhoB

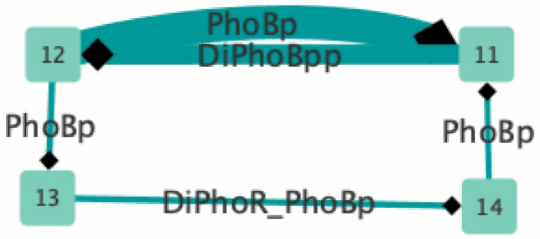
Stochastic Flux Analysis – first 15 minutes

flux weight > **20%**

1	DiPhoR → DiPhoRp
2	DiPhoRp → DiPhoR
3	DiPhoRp → DiPhoRpp
4	DiPhoRpp → DiPhoRp
5	DiPhoRpp + PhoB → DiPhoRpp_PhoB
6	DiPhoRpp_PhoB → DiPhoRpp + PhoB
7	DiPhoRpp_PhoB → DiPhoRp + PhoBp
8	DiPhoRp + PhoB → DiPhoRp_PhoB
9	DiPhoRp_PhoB → DiPhoRp + PhoB
10	DiPhoRp_PhoB → DiPhoR + PhoBp
11	PhoBp + PhoBp → DiPhoBpp
12	DiPhoBpp → PhoBp + PhoBp
13	DiPhoR + PhoBp → DiPhoR_PhoBp
14	DiPhoR_PhoBp → DiPhoR + PhoBp
15	DiPhoR_PhoBp → DiPhoR + PhoB

Stochastic Flux Analysis – first 15 minutes

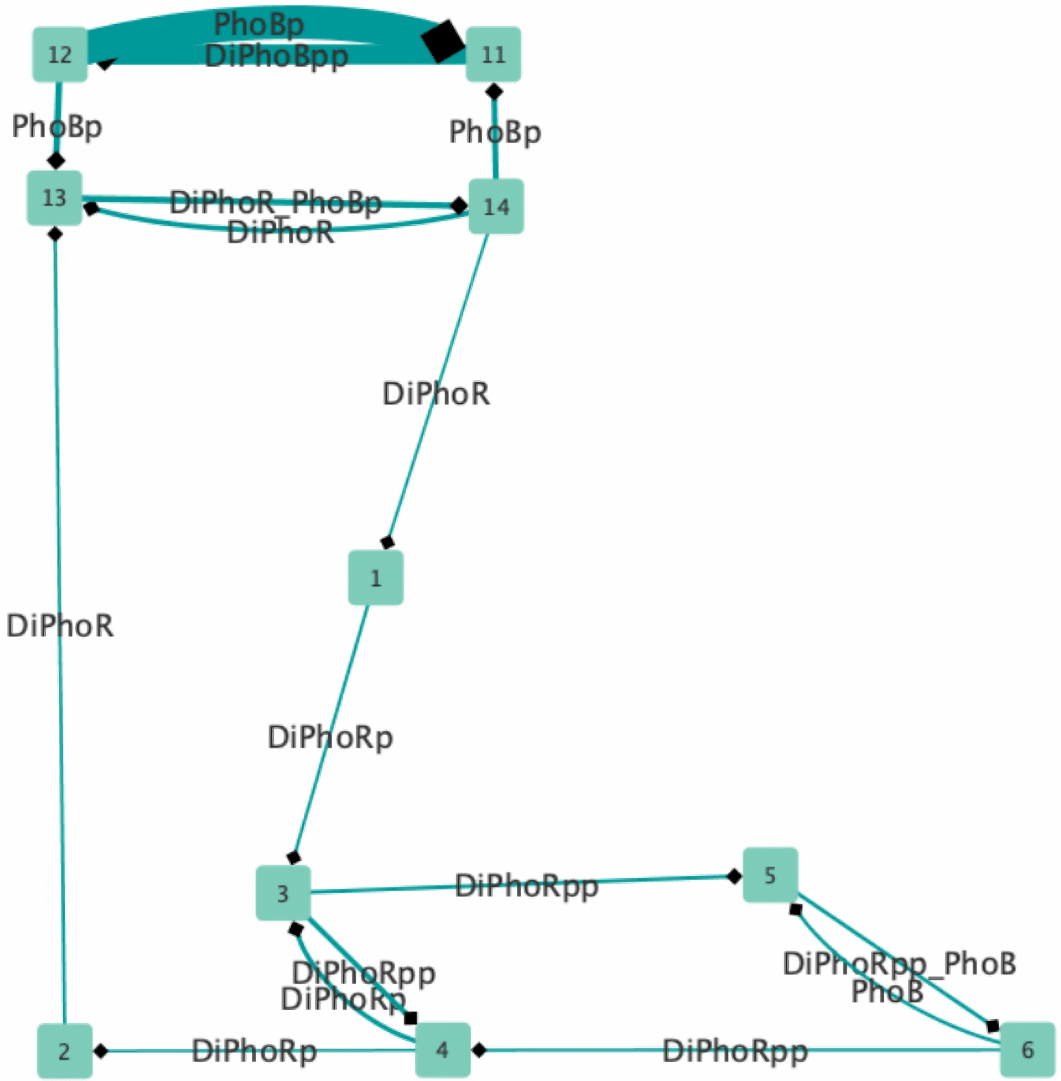
flux weight > **10%**



1	$\mathrm{DiPhoR} \rightarrow \mathrm{DiPhoRp}$
2	$\mathrm{DiPhoRp} \rightarrow \mathrm{DiPhoR}$
3	$\mathrm{DiPhoRp} \rightarrow \mathrm{DiPhoRpp}$
4	$\mathrm{DiPhoRpp} \rightarrow \mathrm{DiPhoRp}$
5	$\mathrm{DiPhoRpp} + \mathrm{PhoB} \rightarrow \mathrm{DiPhoRpp_PhoB}$
6	$\mathrm{DiPhoRpp_PhoB} \rightarrow \mathrm{DiPhoRpp} + \mathrm{PhoB}$
7	$\mathrm{DiPhoRpp_PhoB} \rightarrow \mathrm{DiPhoRp} + \mathrm{PhoBp}$
8	$\mathrm{DiPhoRp} + \mathrm{PhoB} \rightarrow \mathrm{DiPhoRp_PhoB}$
9	$\mathrm{DiPhoRp_PhoB} \rightarrow \mathrm{DiPhoRp} + \mathrm{PhoB}$
10	$\mathrm{DiPhoRp_PhoB} \rightarrow \mathrm{DiPhoR} + \mathrm{PhoBp}$
11	$\mathrm{PhoBp} + \mathrm{PhoBp} \rightarrow \mathrm{DiPhoBpp}$
12	$\mathrm{DiPhoBpp} \rightarrow \mathrm{PhoBp} + \mathrm{PhoBp}$
13	$\mathrm{DiPhoR} + \mathrm{PhoBp} \rightarrow \mathrm{DiPhoR_PhoBp}$
14	$\mathrm{DiPhoR_PhoBp} \rightarrow \mathrm{DiPhoR} + \mathrm{PhoBp}$
15	$\mathrm{DiPhoR_PhoBp} \rightarrow \mathrm{DiPhoR} + \mathrm{PhoB}$

Stochastic Flux Analysis – first 15 minutes

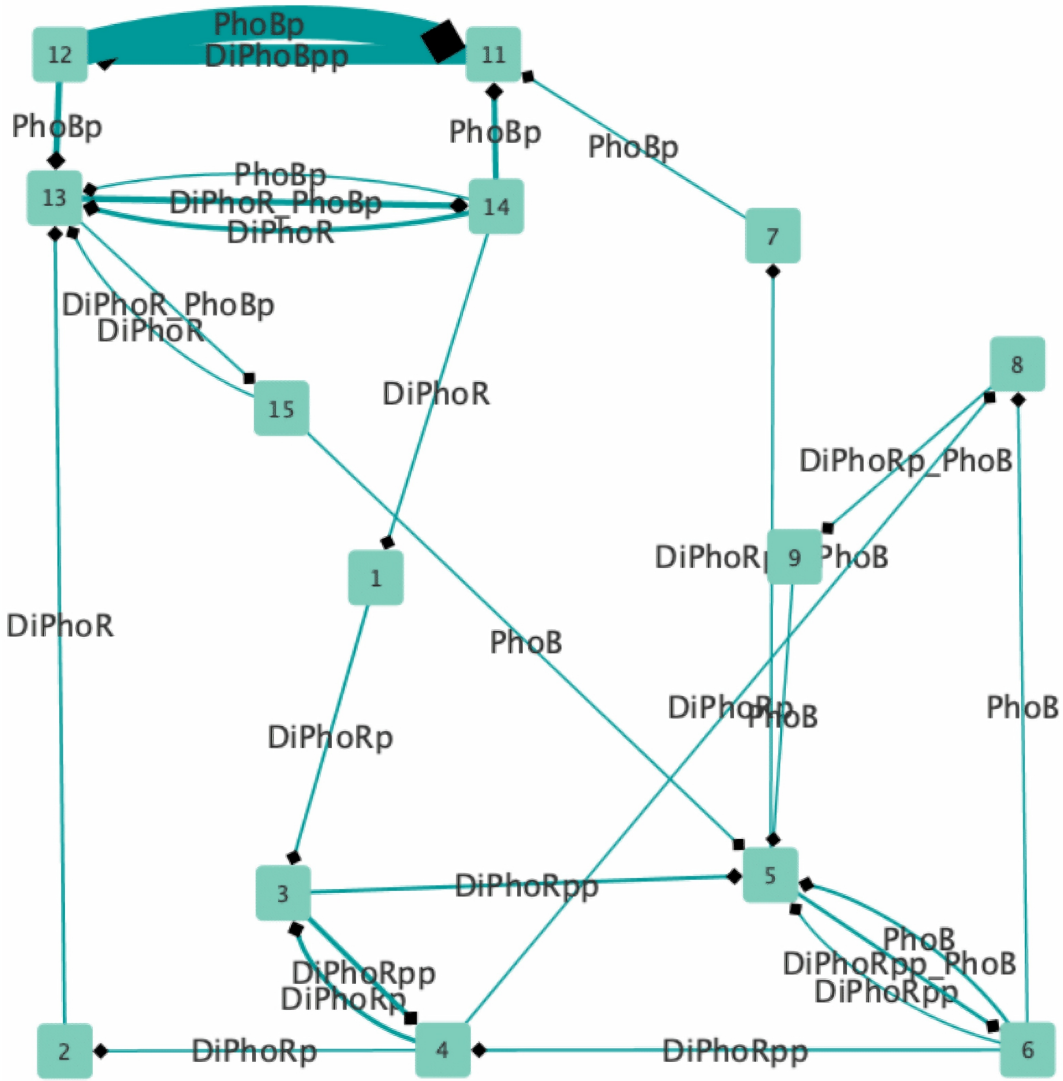
flux weight > **2%**



1. DiPhoR → DiPhoRp
2. DiPhoRp → DiPhoR
3. DiPhoRp → DiPhoRpp
4. DiPhoRpp → DiPhoRp
5. DiPhoRpp + PhoB → DiPhoRpp_PhoB
6. DiPhoRpp_PhoB → DiPhoRpp + PhoB
7. DiPhoRpp_PhoB → DiPhoRp + PhoBp
8. DiPhoRp + PhoB → DiPhoRp_PhoB
9. DiPhoRp_PhoB → DiPhoRp + PhoB
10. DiPhoRp_PhoB → DiPhoR + PhoBp
11. PhoBp + PhoBp → DiPhoBpp
12. DiPhoBpp → PhoBp + PhoBp
13. DiPhoR + PhoBp → DiPhoR_PhoBp
14. DiPhoR_PhoBp → DiPhoR + PhoBp
15. DiPhoR_PhoBp → DiPhoR + PhoB

Stochastic Flux Analysis – first 15 minutes

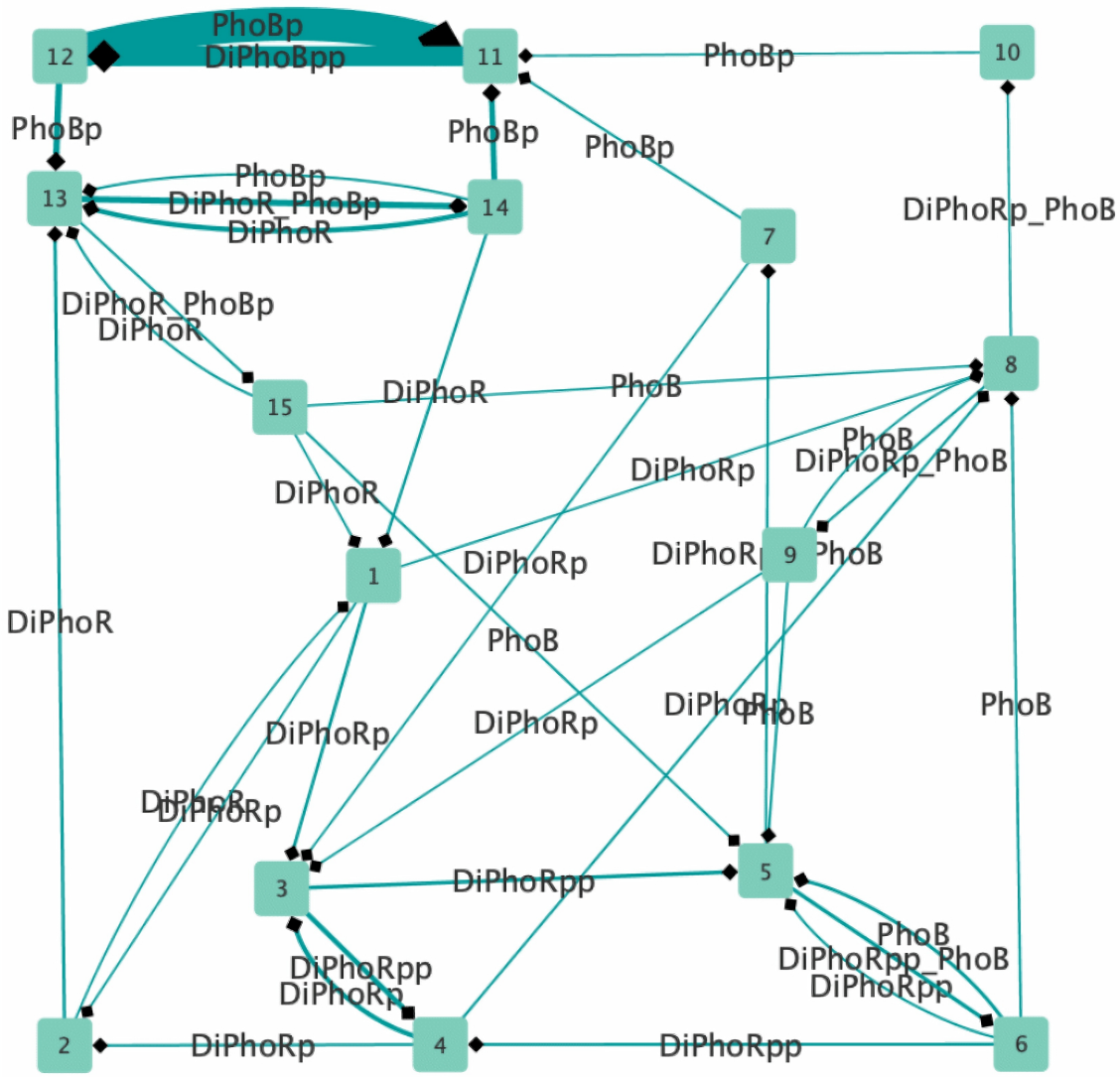
flux weight > 1%



1. DiPhoR → DiPhoRp
2. DiPhoRp → DiPhoR
3. DiPhoRp → DiPhoRpp
4. DiPhoRpp → DiPhoRp
5. DiPhoRpp + PhoB → DiPhoRpp_PhoB
6. DiPhoRpp_PhoB → DiPhoRpp + PhoB
7. DiPhoRpp_PhoB → DiPhoRp + PhoBp
8. DiPhoRp + PhoB → DiPhoRp_PhoB
9. DiPhoRp_PhoB → DiPhoRp + PhoB
10. DiPhoRp_PhoB → DiPhoR + PhoBp
11. PhoBp + PhoBp → DiPhoBpp
12. DiPhoBpp → PhoBp + PhoBp
13. DiPhoR + PhoBp → DiPhoR_PhoBp
14. DiPhoR_PhoBp → DiPhoR + PhoBp
15. DiPhoR_PhoBp → DiPhoR + PhoB

Stochastic Flux Analysis – first 15 minutes

flux weight > **0.9%**



1. DiPhoR → DiPhoRp
2. DiPhoRp → DiPhoR
3. DiPhoRp → DiPhoRpp
4. DiPhoRpp → DiPhoRp
5. DiPhoRpp + PhoB → DiPhoRpp_PhoB
6. DiPhoRpp_PhoB → DiPhoRpp + PhoB
7. DiPhoRpp_PhoB → DiPhoRp + PhoBp
8. DiPhoRp + PhoB → DiPhoRp_PhoB
9. DiPhoRp_PhoB → DiPhoRp + PhoB
10. DiPhoRp_PhoB → DiPhoR + PhoBp
11. PhoBp + PhoBp → DiPhoBpp
12. DiPhoBpp → PhoBp + PhoBp
13. DiPhoR + PhoBp → DiPhoR_PhoBp
14. DiPhoR_PhoBp → DiPhoR + PhoBp
15. DiPhoR_PhoBp → DiPhoR + PhoB

Summary

- A computational model of TCS and its experimental validation for mechanisms of auto-regulation in *E. coli.*
- Predictions for the design of a spectrum of synthetic promoters for gene expression for enhanced P_i intake.
- Implications in biotechnology applications such as biosensors or wastewater treatment systems.
- Quantification of active components of the metabolism at phases of the system via stochastic flux analysis.

Uluşeker et al., *A Dynamic Model of the Phosphate Response System with Synthetic Promoters in E. coli*. **ECAL** 2017

Uluşeker et al. Dynamic mechanisms of phosphate response with synthetic promoters in *Escherichia coli,* ***Scientific Reports****, 2019, in press*

Ozan Kahramanoğulları, Quantifying Information Flow in Chemical Reaction Networks, **AICoB**, 2017.

Ozan Kahramanoğulları, James Lynch, Stochastic Flux Analysis of Chemical Reaction Networks, **BMC Systems Biology**, 2013.

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